Chapter 3
pn-Junctions

pn-junctions are a basic element of nearly all power devices. They are formed when the type of conductivity changes from p to n-type within the same crystal. pn-junctions are rectifying, they conduct current only in one direction of the applied voltage, called forward direction, whereas in the opposite direction, the blocking direction, the current is extremely small. The rectification of pn-junctions has been described first theoretically by Davidov [Dav38]. After some further work (see [Sho49]) the pn theory was developed in a more complete form by Shockley [Sho49, Sho50]. Together with the invention of the transistor this was a starting point of the enormous development of the semiconductor industry till now. The operation of the metal-semiconductor rectifier was known already at that time [Sch38, Sch39]. Today, even the early poly-crystal rectifiers are thought to function mostly by a pn-junction.

The rectifying effect can be simply understood qualitatively (see Fig. 3.1). If a positive voltage is applied to the p-region with respect to the n-region, then the free holes in the p-region and the free electrons in the n-region are driven towards the junction and are injected partly into the opposite region as excess minority carriers. Since there is no lack of carriers for the current flow, the pn-junction is conducting in this bias condition. If the voltage at the p-region is negative with respect to the n-region, then both types of majority carriers are withdrawn from the junction and cannot be supplied from the adjacent region of opposite conductivity, except for the few equilibrium minority carriers there. Hence only a very small current can flow, the pn-junction is biased in the reverse or blocking direction.

3.1 The pn-Junction in Thermal Equilibrium

First we consider a pn-junction in thermodynamic equilibrium, the case of zero external voltage and current. The formulae for this case can be transferred to a large extent to the case of applied voltage and hence are useful also for the I-V

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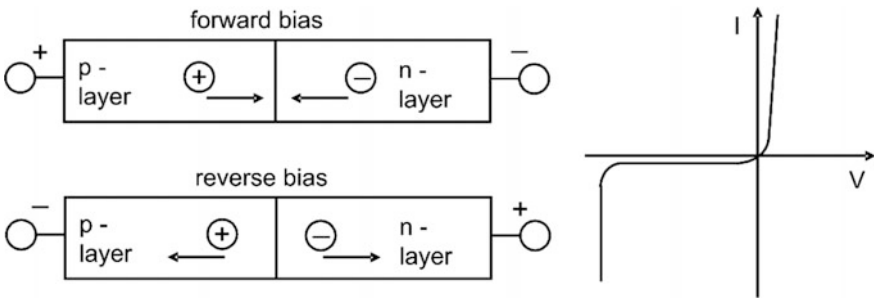


Fig. 3.1 pn-junction in forward- and in blocking direction

characteristics in forward and reverse direction. As discussed previously, the concentration of free holes in the bulk of the p-region is equal to the concentration of ionized acceptors, likewise deep in the n region the electron density is given by the donor concentration, thus the charge of the impurities is neutralized by the carriers. This is not the case, however, near the transition between the p and n region, as is indicated in Fig. 3.2. Here the hole density in the p region has a steep slope $-dp/dx$ resulting in a diffusion of holes towards the n-region. Since the fixed acceptors stay behind without compensation, a negative space charge arises in the p region near the junction. From the n-region, in the same way electrons diffuse towards the p region, so that a positive space charge of uncompensated donors remains in the n region near the junction. Between both space charges an electric field is built up, which drives the holes towards the p region and the electrons towards the n region, i.e. in both cases in opposite direction to the particle diffusion currents. The thermal equilibrium is reached when the field current compensates the diffusion current both for electrons and holes. The field over the space charge region results in a built-in voltage $V_{bi} = -\int_{x_p}^{x_n} \mathsf{E}\, dx$, where x_p, x_n are the boundaries of the space charge layer in the p and n region, respectively. The built-in potential holds the electrons in the n region and the holes in the p region. V_{bi} is often called also 'diffusion voltage', because its primary cause is diffusion.

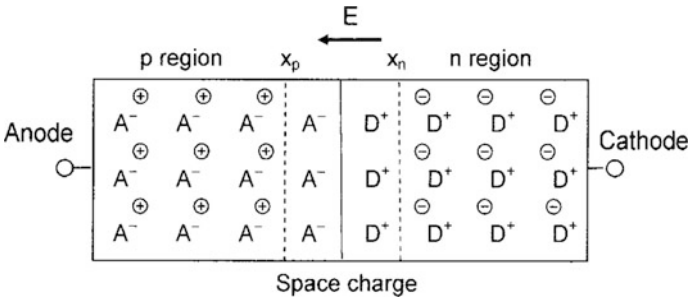


Fig. 3.2 pn-junction in thermal equilibrium

For the built-in voltage a simple general relationship can be derived. In Sect. 2.6, the Einstein relation (2.44) has been derived from the Boltzmann distribution (2.45). Inversely, using the Einstein relation in the current equations (2.43a), (2.43b)

$$j_n = q \cdot \mu_n \left(n \cdot \mathsf{E} + \frac{kT}{q} \frac{dn}{dx} \right) \tag{3.1}$$

$$j_p = q \cdot \mu_p \left(p \cdot \mathsf{E} - \frac{kT}{q} \frac{dp}{dx} \right) \tag{3.2}$$

and setting $j_p = 0$ in (3.2) one obtains the Boltzmann distribution by integrating the bracket term:

$$p(x) = p(x_p) \cdot e^{-\frac{q \cdot V(x)}{kT}}. \tag{3.3}$$

where the potential is given as $V(x) = -\int_{x_p}^{x} \mathsf{E}(x')dx'$. Similarly, one obtains from Eq. (3.1) for the electrons

$$n(x) = n(x_p) \cdot e^{\frac{q \cdot V(x)}{kT}} \tag{3.4}$$

By these equations the position dependent carrier densities in the space charge region are connected with the potential, which however is also not yet known as function of x. In Sect. 2.3 the relationship $n \cdot p = n_i^2$ has been derived for thermal equilibrium, without considering the presence of an electrical field. We see now that this relationship follows from (3.3), (3.4) at every point also in the space charge region where n and p are varying with x:

$$n(x) \cdot p(x) = n(x_p) \cdot p(x_p) = n_i^2 \tag{3.5}$$

The built-in voltage V_{bi} is obtained from (3.3) or (3.4) as $V_{bi} = V(x_n)$:

$$V_{bi} = \frac{kT}{q} \ln \frac{p(x_p)}{p(x_n)} = \frac{kT}{q} \ln \frac{p(x_p) \cdot n(x_n)}{n_i^2} \tag{3.6}$$

$$\cong \frac{kT}{q} \ln \frac{N_A(x_p) \cdot N_D(x_n)}{n_i^2} \tag{3.6a}$$

In the approximation (3.6a) complete ionization of the dopants is assumed together with neutrality at the boundaries of the space charge layer (see Sect. 2.5). According to this equation the built-in voltage is given by the doping densities at the boundaries of the space charge region.

The space dependence of the potential and carrier concentrations and the extension of the space charge layer into the p and n region can be calculated from the Poisson equation (2.111). Assuming as before complete ionization of the impurities and using (3.3), (3.4) one has for arbitrary doping profiles

$$\begin{aligned}\frac{d^2V}{dx^2} &= -\frac{\rho}{\varepsilon} = \frac{q}{\varepsilon}\left(n - p + N_{A,tot}(x) - N_{D,tot}(x)\right) \\ &= \frac{q}{\varepsilon}\left(n(x_p)\cdot e^{\frac{q\cdot V}{kT}} - p(x_p)\cdot e^{-\frac{q\cdot V}{kT}} - N(x)\right)\end{aligned} \tag{3.7}$$

where $N(x) \equiv N_{D,tot} - N_{A,tot}$. The donor and acceptor concentrations are indicated now additionally by a subscript '*tot*' to distinguish them from the net doping densities $N_A(x) = N_{A,tot}(x) - N_{D,tot}(x)$ in the p-region and $N_D(x) = N_{D,tot} - N_{A,tot}$ in the n-region which appear in Eq. (3.6a). The carrier concentrations at the p sided boundary of the space charge region are then $p_p(x_p) = N_A(x_p)$, $n(x_p) = n_i^2/N_A(x_p)$. Generally, the ordinary differential Eq. (3.7) has to be solved numerically, but for abrupt step junctions $V(x)$ can be expressed analytically by an integral (see Sect. 3.1.1). As will be seen, however, an approximate simpler calculation is often sufficient and even more useful. The exact solution of (3.7) will be used to correct the approximate formulae where it is necessary.

As follows from (3.6a) with $N_A(x_p)$, $N_D(x_n) >> n_i$, the built-in voltage is much larger than the thermal voltage kT/q. Hence a variation of the potential by a few kT/q at the boundaries of the space charge region towards the interior, connected with a decrease of the majority carrier concentrations towards zero, needs only a small distance. It seems reasonable therefore to neglect the carrier concentrations in the space charge region. Then (3.7) reduces to

$$\frac{d^2V}{dx^2} = -\frac{q}{\varepsilon}N(x) \tag{3.8}$$

This approach considering the space charge region as depleted from carriers is called depletion approximation, it will be used extensively in what follows.

3.1.1 The Abrupt Step Junction

The abrupt pn-junction or abrupt step junction is defined by a sharp, step-like doping transition between the p and n region and homogeneous doping within each of the two regions. Since N_A and N_D are independent of the positions x_p and x_n of the boundaries of the space charge layer, the built-in voltage is immediately known by (3.6a). In Fig. 3.3, the built-in voltage of abrupt pn-junctions in silicon is plotted for two temperatures versus the doping concentration N_D of the n region assuming a

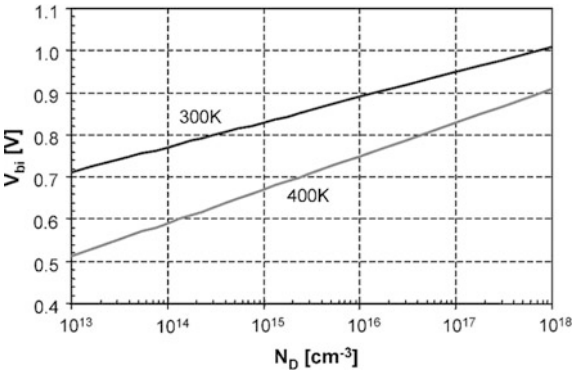


Fig. 3.3 Built-in voltage of abrupt pn-junctions in silicon as function of the doping concentration of the weakly doped side (N_D) for a fixed doping of 1×10^{19} cm^{-3} (N_A) of the highly doped region

fixed doping $N_A = 1 \times 10^{19}$ cm^{-3} of the p region. At 300 K, V_{bi} increases from 0.705 to 1.00 V in the range 1×10^{13}–1×10^{18} cm^{-3}, at 400 K from 0.508 to 0.905 V. The decrease of V_{bi} with increasing T is due to the strong increase of n_i. The decrease withtemperature is approximately linear, because the pre-exponential factor of n_i^2 in Eq. (2.6), $N_c \cdot N_v$, is larger than $N_D \cdot N_A$ in the application range of the formula and its logarithm nearly constant like E_g. At high doping concentrations both of the p and the n region the temperature dependence of V_{bi} is only weak.

We will now use the depletion approximation to calculate the potential and carrier concentrations in an abrupt junction as a function of x. Figure 3.4 illustrates the space dependence of (a) the doping and carrier concentrations, (b) the resulting charge density, (c) the electric field in the depletion approximation, (d) the potential and (e) the corresponding band diagram. These qualitative plots are substantiated now by the following calculations.

Placing the origin of x at the transition between the acceptor and donor doping, the metallurgical junction, the doping profile $N(x)$ is given by

$$\begin{aligned} N(x) &= -N_A = \text{const} \quad \text{for } x < 0, \\ N(x) &= +N_D = \text{const} \quad \text{for } x \geq 0 \end{aligned} \tag{3.9}$$

In the region with acceptor doping the Poisson equation

$$\frac{d^2V}{dx^2} = \frac{q}{\varepsilon} N_A \tag{3.10}$$

results in the field (see Fig. 3.4)

$$-\frac{dV}{dx} = \mathsf{E}(x) = -\frac{q}{\varepsilon} N_A \cdot (x - x_p), \quad \text{for } x_p \leq x \leq 0 \tag{3.11}$$

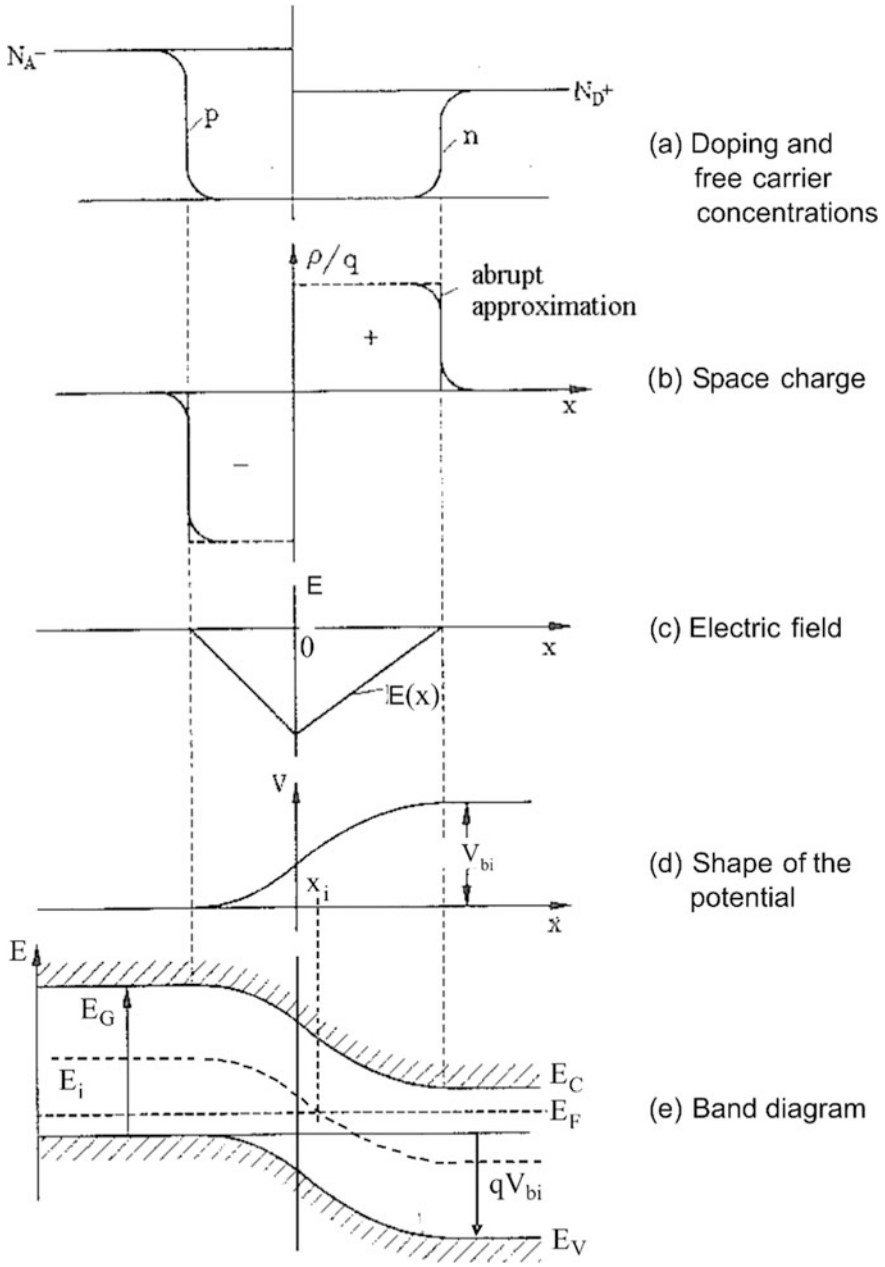


Fig. 3.4 Abrupt pn junction in the depletion approximation

For the metallurgical n-region, integration of the Poisson equation $d^2V/dx^2 = -q/\varepsilon N_D$ yields

$$-\frac{dV}{dx} = \mathsf{E}(x) = -\frac{q}{\varepsilon} N_D \cdot (x_n - x) \quad \text{for } 0 \le x \le x_n \tag{3.12}$$

The continuity of E at $x = 0$ requires

$$N_A \cdot x_p = -N_D \cdot x_n \tag{3.13}$$

meaning that the charges on both sides of the junction are oppositely equal. Integration of (3.11), (3.12) leads to

$$V(x) = \frac{q}{2\varepsilon} \cdot N_A \cdot (x - x_p)^2 \quad \text{for } x_p < x \le 0 \tag{3.14}$$

and

$$V(x) = \frac{q}{2\varepsilon} \left[-N_D \cdot (x_n - x)^2 + N_D x_n^2 + N_A x_p^2 \right] \quad \text{for} \quad 0 \le x < x_n \tag{3.15}$$

where the constant term in (3.15) is chosen to get continuity at $x = 0$. As is shown by (3.11), (3.12) and Fig. 3.4c, the field strength in each region is linear in x, the potential has a parabolic course. The ratio of the potential increase in the acceptor region to that in the donor region is obtained from (3.13) to (3.15) as

$$\frac{V_P}{V_N} = \frac{N_A \cdot x_p^2}{N_D \cdot x_n^2} = \frac{|x_p|}{x_n} = \frac{N_D}{N_A} \tag{3.16}$$

The penetration depths $-x_p$ and x_n can now be determined equating $V(x_n)$ to the built-in voltage as given by (3.6a). From (3.15), (3.13) one obtains

$$V_{bi} = \frac{q}{2\varepsilon} \cdot \left(N_D \cdot x_n^2 + N_A x_p^2 \right) = \frac{q}{2\varepsilon} \cdot \left(N_D \cdot x_n^2 + \frac{N_D^2}{N_A} x_n^2 \right) \tag{3.17}$$

$$x_n = \sqrt{\frac{2\varepsilon}{q} \cdot \frac{N_A / N_D}{(N_A + N_D)} \cdot V_{bi}}, \quad |x_p| = \frac{N_D}{N_A} \cdot x_n \tag{3.18}$$

The total thickness of the space charge layer is

$$w_{sc} = x_n + |x_p| = \sqrt{\frac{2\varepsilon}{q} \cdot \frac{N_A + N_D}{N_A N_D} \cdot V_{bi}} \tag{3.19}$$

The maximum absolute field strength $\mathsf{E}_m = |\mathsf{E}(0)|$ is obtained from (3.12), (3.18) as:

$$\mathsf{E}_m = \sqrt{\frac{2q}{\varepsilon} \cdot \frac{N_A N_D}{N_A + N_D} \cdot V_{bi}} = \frac{2 \cdot V_{bi}}{w_{sc}} \tag{3.20}$$

For asymmetric junctions with very different concentrations N_A, N_D, as found often in devices, the formulae simplify. Referring in the notation to a p^+n junction with $N_A \gg N_D$, Eq. (3.19) turns into

$$w_{sc} \approx x_n \approx \sqrt{\frac{2\varepsilon \cdot V_{bi}}{q \cdot N_D}}. \tag{3.19a}$$

whereas (3.20) yields

$$\mathsf{E}_m \approx \sqrt{\frac{2q}{\varepsilon} \cdot N_D \cdot V_{bi}} \tag{3.20a}$$

In the energy band diagram of Fig. 3.4e, the band edges vary inversely to the potential because the conduction band edge E_c represents the potential *energy* $-qV(x)$ of the electrons. As a general equilibrium condition the Fermi level E_F is constant across the pn-junction, which follows also from Eqs. (2.4) and (3.4). The built-in voltage multiplied with the elementary charge q is represented in the band diagram by the entire change of the band edges or of the intrinsic level E_i against the Fermi level. The point x_i where the intrinsic level crosses the Fermi level divides the region where $p > n$ from the region with $n > p$. This point differs generally from the metallurgical junction as is indicated in Fig. 3.4e. Instead of the above calculated voltage parts V_P, V_N in the metallurgical p and n region, the built-in voltage can be divided also into the potential difference $\Delta V_{p\text{-}i}$ between the neutral p region and the intrinsic point x_i on one side and the potential difference $\Delta V_{i\text{-}n}$ between the intrinsic point and the neutral n region on the other. Using (2.11), (2.10) these parts are obtained as

$$\begin{aligned} \Delta V_{p-i} &= \frac{1}{q}(E_i(x_p) - E_F) = \frac{kT}{q}\ln\frac{p(x_p)}{n_i} \cong \frac{kT}{q}\ln\frac{N_A}{n_i} \\ \Delta V_{i-n} &= \frac{1}{q}(E_F - E_i(x_n)) = \frac{kT}{q}\ln\frac{n(x_n)}{n_i} \cong \frac{kT}{q}\ln\frac{N_D}{n_i} \end{aligned} \tag{3.21}$$

If $N_A > N_D$ the voltage $\Delta V_{p\text{-}i}$ is larger than $\Delta V_{i\text{-}n}$, whereas $V_P < V_N$.

As numerical example we consider a pn-junction in silicon with $N_A = 2 \times 10^{15}$ cm^{-3}, $N_D = 1 \times 10^{15}$ cm^{-3}. The built-in voltage is in this case 0.604 V. Using this, Eqs. (3.18) with the permittivity of silicon $\varepsilon_r = 11.7$ yield the penetration depths $x_n = 0.725$ µm, $|x_p| = 0.363$ µm. The maximum value of the field according to (3.20) is $\mathsf{E}_m = 1.12 \times 10^4$ V/cm. The parts of the built-in voltage in the metallurgical acceptor and donor region are $V_P = N_D/(N_A + N_D) \cdot V_{bi} = V_{bi}/3$

= 0.201 V, $V_N = 2 \cdot V_P = 0.402$ V. In contrast to this, Eq. (3.20) yields for the voltage increase in the region with $p > n_i$ a value $\Delta V_{p\text{-}I} = 0.311$ V, while the voltage in the region with $n > n_i$ is $\Delta V_{i\text{-}n} = 0.296$ V. The intrinsic point is shifted into the region with lower doping. Numerically (3.15) together with (3.21) yields $x_i = 0.106$ µm.

As will be seen, Eqs. (3.19a), (3.20a) have only a very limited applicability just for well abrupt asymmetrical junctions, whereas for diffused junctions they are very useful. Their accuracy can be tested by comparing with results following from the exact Poisson Eq. (3.7). Integrating 3.7 once,[1] the following relationship between the potential parts V_P, V_N in the acceptor and donor doped region, respectively, is obtained instead of (3.16):

$$\frac{V_P - kT/q}{V_N - kT/q} = \frac{N_D}{N_A} \tag{3.22}$$

Together with $V_P + V_N = V_{bi}$ it follows explicitly

$$\begin{aligned} V_P &= \frac{N_A - N_D}{N_A + N_D} \cdot \frac{kT}{q} + \frac{N_D}{N_A + N_D} \cdot V_{bi} \\ V_N &= V_{bi} - V_P \end{aligned} \tag{3.23}$$

For the maximal field the exact calculation yields:

$$\mathsf{E}_m = \sqrt{\frac{2kT}{\varepsilon} \cdot \left\{ N_A \cdot e^{-\frac{qV_P}{kT}} + N_D \cdot \left(\frac{qV_N}{kT} + e^{-\frac{qV_N}{kT}} - 1 \right) \right\}} \tag{3.24}$$

Equations (3.22)–(3.24) hold for arbitrary concentrations N_A, N_D. If V_P, $V_N >> kT/q$, (3.22) turns into (3.16), and in (3.24) only the linear term in V_N remains which with (3.23) leads to (3.20). For the doping concentrations this requires that N_A/N_D as well as N_D/N_A are small against qV_{bi}/kT. Since on the other hand $N_A/N_D \gg 1$ is required for (3.19a), (3.20a), these equations are roughly valid only in a small range around $N_A/N_D = 5$ in Si. In contrast to (3.20a), Eq. (3.24) yields an increase of E_m with N_A for constant N_D. For a *very* asymmetrical p^+n junction defined by $N_A/N_D \gg qV_{bi}/kT$ the voltage V_P tends to kT/q according to (3.23) (instead of zero after (3.16)), and now the N_D-term in (3.24) is negligible. Hence (3.24) turns into

$$\mathsf{E}_m = \sqrt{\frac{2kT}{\varepsilon} N_A/e} \quad \text{for } N_A/N_D \gg qV_{bi}/kT \tag{3.24a}$$

[1] After multiplication with $2 \cdot dV/dx$, Eq. (3.7) can be integrated analytically to obtain the field $\mathsf{E}(x)$ as function of $V(x)$ in both regions. From the continuity of potential and field at the metallurgical junction the potential parts V_P, V_N are then determined.

While according to the 'classical' formula (3.20a) the maximum field is determined by the doping concentration of the weakly doped region (and the built-in voltage), Eq. (3.24a) says that it is given only by the doping density of the highly doped region. Also numerically the results are extremely different: For the example $N_A = 5 \times 10^{18}$ cm^{-3}, $N_D = 1 \times 10^{14}$ cm^{-3} in silicon, one obtains from (3.20a) $\mathsf{E}_\mathrm{m} = 4.82 \times 10^3$ V/cm, whereas Eq. (3.24a) and also (3.24) result in $\mathsf{E}_\mathrm{m} = 1.21 \times 10^5$ V/cm.

The cause of this strong discrepancy becomes evident from Fig. 3.5, which shows the carrier distributions in a p$^+$n junction as calculated by twice integration of Eq. (3.7). It is seen that the hole concentration of the acceptor doping is not restricted to the N_A region but reaches beyond the metallurgical junction into the N_D region. In the first range of the latter the hole concentration is much higher than the doping density N_D and even the integrated hole charge in the N_D region is large compared with the integral charge of ionized donors in the space charge region. This explains why the field at the metallurgical junction is independent of N_D according to (3.24a). As will be shown in Sect. 3.5, the charge of mobile carriers in the space charge region has a strong influence on the capacitance of the junction.

The assumed exact abruptness is realized mostly by pn junctions in wide-gap semiconductors like SiC, where the doping profiles are not smoothed out by diffusion. In silicon, pn-junctions made by low-temperature epitaxy are very abrupt. For diffused junctions Eq. (3.24) is usually not applicable, because the doping concentration of the highly doped region decreases more slowly than the carrier concentration. The positive charge in the space charge region of a p$^+$n junction is then given solely by the donor doping. Hence Eqs. (3.19a), (3.20a) are approximately applicable for a wide class of diffused p$^+$n junctions, whereas they are not valid for abrupt junctions with $N_A/N_D \gtrsim qV_{bi}/kT$. This holds especially in the extended form which includes an applied voltage (see Sect. 3.3).

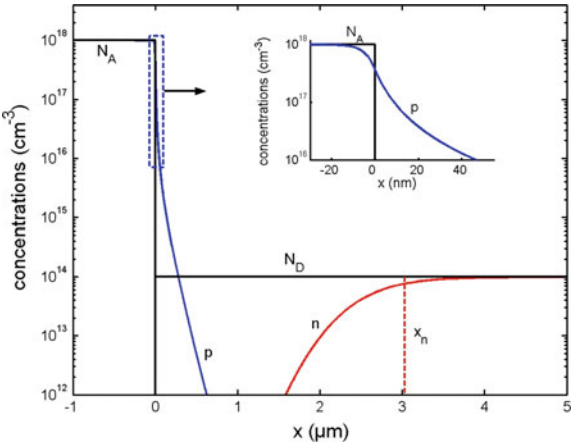


Fig. 3.5 Carrier distribution in an abrupt p$^+$n-junction. x_n is the boundary of the space charge region in the depletion approximation

3.1.2 Graded Junctions

Often the transition between the acceptor and donor doping is not abrupt, but occurs with a rather low gradient which then becomes significant for the properties of the junction. Particularly this is the case for pn-junctions produced by deep diffusion of impurities (for details of this technique see Sect. 4.4). If the diffused dopant is of the opposite type and has a higher surface concentration than the doping of the wafer, a pn-junction is formed at the point where the diffused impurity just compensates the wafer doping. The net doping density, $N(x) = N_D - N_A(x)$ for an acceptor diffusion, changes its sign at the junction. If the gradient of $N(x)$ is not too small, a significant space charge builds up as for abrupt junctions. In Fig. 3.6 the profile of a diffused acceptor with surface concentration 1×10^{18} cm^{-3} added to a homogeneous donor doping with density 1×10^{14} cm^{-3} is shown together with the resulting absolute net doping density and the electron and hole concentration calculated numerically for silicon. Near the pn-junction, the carrier concentrations are much smaller than the net doping density, why the space charge there is a large. The net doping density shown on an expanded linear scale in the inset varies nearly over the whole space charge layer. The depletion approach neglecting the carrier charge in the space charge region can be used also in this case as an approximate basis. Sufficiently near to the junction, the net doping concentration can be approximated by a linear dependence

$$N(x) = a \cdot x, \tag{3.25}$$

using the junction as origin of x. Assuming that this holds over the whole space charge region one obtains the *linearly graded junction* model which seems to be applicable to such doping profiles rather than the abrupt junction model. Under these conditions one obtains from (3.8), (3.25)

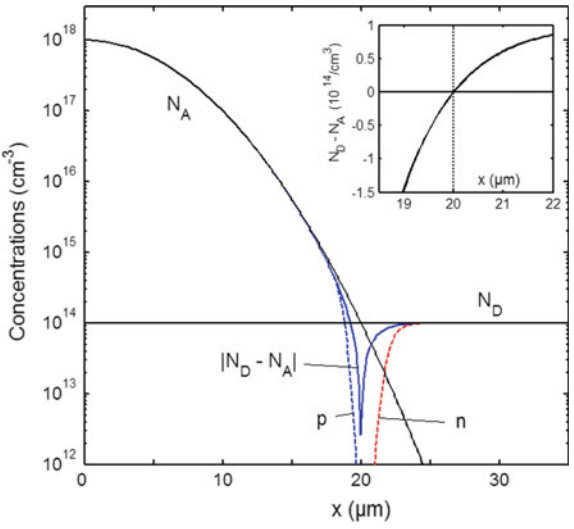


Fig. 3.6 Diffused p^+n-junction in silicon: Doping profile together with hole and electron distribution. A Gaussian function, $N_A \propto \exp(-(x/L_A)^2)$, is assumed as diffusion profile

$$\frac{d^2V}{dx^2} = -\frac{d\mathsf{E}}{dx} = -\frac{q \cdot a}{\varepsilon} \cdot x \tag{3.26}$$

$$\mathsf{E}(x) = \frac{q \cdot a}{2\,\varepsilon}\left(x^2 - w^2\right) \tag{3.27}$$

$$V(x) = \frac{q \cdot a}{2\,\varepsilon} \cdot \left(\left(w^2 \cdot x - \frac{1}{3} \cdot x^3\right)\right. \tag{3.28}$$

where w is the extension of the space charge layer in each of the two regions (half width of the total space charge layer) and the potential at the junction is set to zero.

To calculate the built-in voltage $V_{bi} = V(w) - V(-w)$ one has from (3.28) the following connection between w and V_{bi}:

$$w = \left(\frac{3\,\varepsilon\,V_{bi}}{2\,q\,a}\right)^{1/3} \tag{3.29}$$

On the other hand the built-in voltage is given by (3.6a):

$$\begin{aligned} V_{bi} &= \frac{kT}{q} \cdot \ln\left(\frac{-N(-w) \cdot N(w)}{n_i^2}\right) = \frac{kT}{q}\ln\left(\frac{(a \cdot w)^2}{n_i^2}\right) \\ &= 2\frac{kT}{q}\ln\left(\frac{a \cdot w}{n_i}\right) \end{aligned} \tag{3.30}$$

Inserting (3.29) one obtains an implicit equation from which V_{bi} can be determined by iteration as a function of the doping gradient a. The result for silicon at 300 and 400 K is shown in Fig. 3.7. By comparing with the exact numerical solution of (3.7) it is found [Mol64, Mor60] that the depletion approach assuming abrupt transition between complete depletion and neutrality is in this case only a rough approximation applicable mainly for impurity gradients

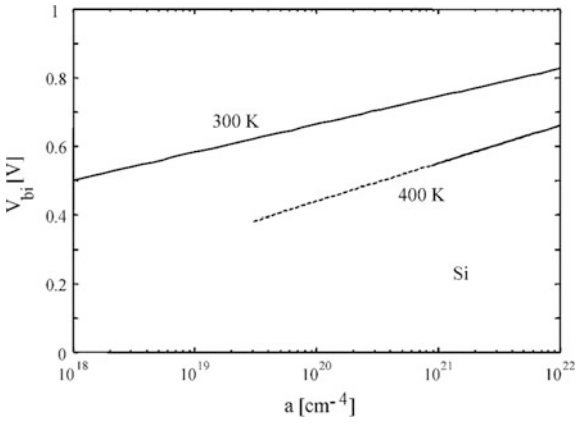


Fig. 3.7 Built-in voltage of a linearly graded junction in Si as function of the doping gradient

$$a \gtrsim 10^4 \frac{2n_i}{L_D} \tag{3.31}$$

where $L_D = \sqrt{\varepsilon kT/(2n_i q^2)}$ is the intrinsic Debye length. For silicon, the right hand side amounts to 7.6×10^{16} cm^{-3} at 300 K, but 8.8×10^{20} cm^{-3} at 400 K. Below this value, the line for 400 K in Fig. 3.7 may be somewhat inexact. The approach can be generalized to the case of an applied external voltage and is used to calculate the capacitance of graded junctions. For these topics we refer to [Mor60, Mol64].

Although far away from a junction and for a doping concentration large against n_i the semiconductor is essentially neutral, a spatially varying doping concentration produces a built-in field and built-in potential also here. Since the electron and hole currents are zero in thermal equilibrium, a field is necessary to compensate the diffusion current arising from the concentration gradient. From Eq. (3.2) one obtains replacing the hole density by the net acceptor concentration

$$\mathsf{E}(x) = \frac{kT}{q}\frac{d}{dx}\ln p = \frac{kT}{q}\frac{d}{dx}\ln N_A \tag{3.32}$$

$$V(x) - V(x_0) = -\frac{kT}{q}\ln\frac{N_A(x)}{N_A(x_0)} \tag{3.33}$$

For n-regions the sign is opposite. For an exponential profile $N(x) \sim e^{x/\lambda}$ the assumed neutrality holds exactly, since the built-in field is constant according to (3.32) and hence the space charge $\rho = \varepsilon\, dE/dx = 0$. The built-in potential of (quasi-) neutral diffused regions is mostly quite appreciable. In the case of Fig. 3.6, the potential difference between the surface of the p-region and the boundary of the space charge region, where $N_A = 3 \times 10^{14}$ cm^{-3}, amounts to 0.28 V at 400 K according to (3.33). If also a diffused n-region follows on the homogeneous n-base as in a pin diode, the whole built-in potential of the quasi-neutral regions comes near to that of the space charge region. For device characteristics, however, the built-in potential of the neutral regions is little significant, because it stays (nearly) constant if an external voltage is applied, whereas the voltage across the space charge region varies essentially, as will be seen. If not indicated else, the term "built-in voltage" refers therefore always to that of the space charge region.

The question arises, why does the built-in voltage not cause a current flow, if the *p* and *n* region are connected externally by a wire. From the constancy of the Fermi level throughout the contacted structure in thermal equilibrium, it follows that contact potentials between the contacted semiconductor regions and the metal exist, and these are in sum oppositely equal to the whole built-in voltage in the semiconductor.

As one of the results of the present section we mention again that the extension of the space charge layer in thermal equilibrium is very small compared with the usual thickness of the (quasi-)neutral p and n regions enclosing it. If an external voltage is applied, the situation can be changed.

3.2 Current-Voltage-Characteristics of the pn-Junction

Now a voltage V is applied to the p region with respect to the n region. If this voltage is positive, it is directed against the built-in voltage V_{bi}. Assuming that the caused current is small and the ohmic voltage drop over the neutral regions can be neglected, the voltage across the space charge region is now

$$\Delta V = V_{bi} - V \tag{3.34}$$

For building up this voltage step, the required charges in the dipole layer of the junction are smaller or larger than without the external voltage depending on the sign of the voltage. Since the charge density on each side is given approximately by the doping density, the thicknesses of the space charge layer in the p and n region decrease or increase. Most important, however, is that the hole concentration in the n region and electron concentration in the p region are raised above the equilibrium minority densities for $V > 0$ and lowered for $V < 0$. This can be concluded from Eqs. (3.3), (3.4) *assuming that the Boltzmann distribution is applicable also to cases away from thermal equilibrium*. This is a basic assumption of the whole device theory, justified by the fact that the deviation from equilibrium is usually weak, i.e. the field and diffusion currents compensate each other largely in the space charge region.

Explicating this, we assume the pn-junction as abrupt and the minority carrier concentrations in the neutral p and n region to be small compared with the doping concentrations. Hence from neutrality one has furthermore $p(x_p) = N_A, n(x_n) = N_D$. Replacing $V(x_n)$ in Eq. (3.3) by (3.34) one obtains then for the hole concentration p_n^* in the neutral n region at the boundary $x = x_n$ to the space charge region

$$p_n^* = N_A \cdot e^{-\frac{q \cdot (V_{bi} - V)}{kT}} \tag{3.35}$$

$$= p_{n0} \cdot e^{\frac{q \cdot V}{kT}} \tag{3.36}$$

Here the equilibrium hole density in the n region is denoted by p_{n0}, and the relationship

$$p_{n0} = N_A e^{-\frac{qV_{bi}}{kT}} = \frac{n_i^2}{N_D} \tag{3.37}$$

was used, which follows from (3.6). For (3.35), (3.36), we have not yet used, that the injection level in the n region is low. This is proposed however now to obtain the electron concentration n_p^* in the neutral p region at the boundary $x = x_p$ to the space charge region. Similarly as (3.36) one obtains:

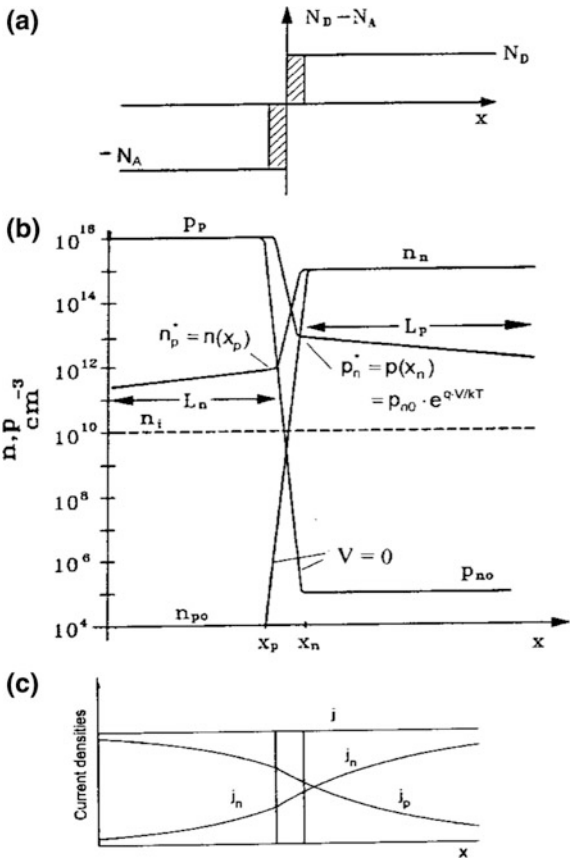


Fig. 3.8 Forward biased pn-junction. **a** Net doping density, **b** Carrier distribution for $V > 0$ and $V = 0$, **c** Hole and electron current densities

$$n_p^* = n_{p0} \cdot e^{\frac{q \cdot V}{kT}} \tag{3.38}$$

where n_{p0} is the equilibrium electron concentration in the neutral p region. The minority concentrations at the boundaries to the space charge region are raised respectively lowered by an exponential factor containing the applied voltage V, the Boltzmann factor. In Fig. 3.8 this is illustrated for forward bias. The applied voltage raises the concentrations p_n^* and n_p^* here by 8 orders of magnitude. On the chosen logarithmic scale the minority carrier concentrations decrease linearly with distance from the space charge layer. The figure anticipates here results of the following calculation.

For further visualization especially of the situation at reverse bias, the minority carrier densities are plotted in Fig. 3.9 on a linear scale. In the picture for reverse bias (lower part of the figure) the minority carrier concentrations at the boundaries to the space charge region are lowered already to zero. This is approached already for reverse voltages higher than a few times the thermal voltage kT/q, as follows

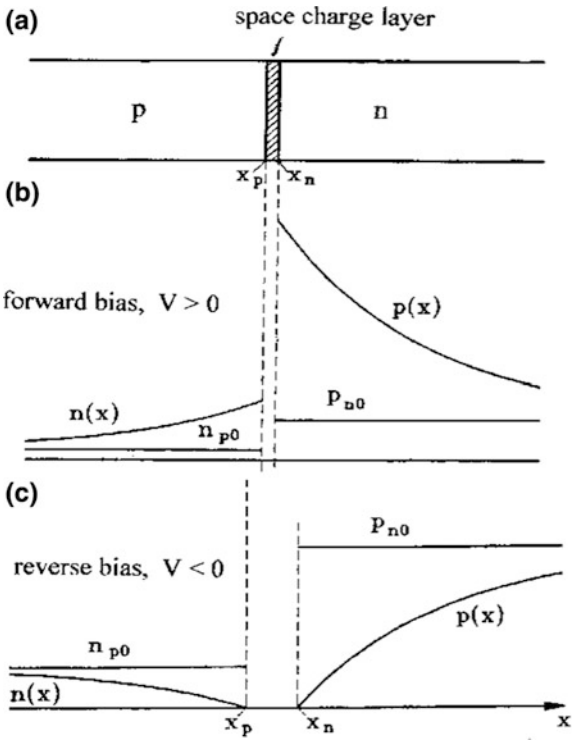


Fig. 3.9 Minority carrier distributions in a pn-junction (**a**) under forward bias (**b**) and reverse bias (**c**) on a linear scale

from (3.36), (3.38). The diffusion of holes out of the n-region and electrons out of the p-region which determines the reverse current can then no longer be enhanced by a further increase of the reverse voltage. Because the equilibrium densities $p_{n0} = n_i^2/N_D$, $n_{p0} = n_i^2/N_A$ are very small and this transfers to the concentration gradients of the minority carriers, the blocking current is also very small.

For the np-product in the space charge region one obtains from Eqs. (3.3), (3.4) together with (3.38)

$$n(x) \cdot p(x) = n(x_p) \cdot p(x_p) = n_{p0} \cdot e^{\frac{qV}{kT}} \cdot p_0 = n_i^2 \cdot e^{\frac{q \cdot V}{kT}} \tag{3.39}$$

As without bias, the np-product in the space charge layer is independent of x, but it is increased or decreased depending on the sign of the voltage V by the exponential voltage factor. The deviation from equilibrium leads to net recombination or generation, respectively.

The I-V characteristic is governed by the minority carrier currents in the neutral regions. To calculate them one uses the continuity equation which for the holes in the n-region according to Eq. (2.103) reads

$$\frac{dj_p}{dx} = -q \cdot R_p = -q \cdot \frac{p - p_{n0}}{\tau_p} \tag{3.40}$$

Here the stationary case is assumed and the excess recombination rate R_p is expressed, according to (2.49), by the minority carrier lifetime and the excess hole concentration. Since the hole density p is assumed small compared with n (low injection level), and additionally the field in the neutral region is small, the field term in the current Eq. (2.43b) can be neglected. Hence we are dealing with the case b of Sect. 2.10.1. Inserting

$$j_p = -q \cdot D_p \frac{dp}{dx} \tag{3.41}$$

into Eq. (3.40) one obtains

$$D_p \cdot \frac{d^2p}{dx^2} = \frac{p - p_{n0}}{\tau_p} \tag{3.42}$$

The solution of this differential equation with the boundary condition $p(x_n) = p_n^*$ is

$$p(x) - p_{n0} = (p_n^* - p_{n0}) \cdot e^{-\frac{x - x_n}{L_p}} \tag{3.43}$$

where L_p is the hole diffusion length:

$$L_p = \sqrt{D_p \cdot \tau_p} \tag{3.44}$$

Inserting p_n^* from (3.36) one has:

$$p(x) - p_{n0} = p_{n0}\left(e^{\frac{qV}{kT}} - 1\right) \cdot e^{-\frac{x - x_n}{L_p}} \tag{3.45}$$

Using this hole distribution in (3.41) the hole current density at $x = x_n$ is obtained as

$$j_p(x_n) = j_{ps} \cdot \left(e^{\frac{q \cdot V}{kT}} - 1\right) \tag{3.46}$$

with

$$j_{ps} = q \cdot p_{n0} \cdot \frac{D_p}{L_p} = q \cdot \frac{n_i^2}{N_D} \cdot \frac{D_p}{L_p} \tag{3.46a}$$

Analogously the electron current density in the p region at $x = x_p$ is given by

$$j_n(x_p) = j_{ns} \cdot \left(e^{\frac{q \cdot V}{kT}} - 1\right) \tag{3.47}$$

with

$$j_{ns} = q \cdot n_{p0} \cdot \frac{D_n}{L_n} = q \cdot \frac{n_i^2}{N_A} \cdot \frac{D_n}{L_n} \tag{3.47a}$$

where L_n is the electron diffusion length:

$$L_n = \sqrt{D_n \cdot \tau_n} \tag{3.48}$$

As is seen, the minority carrier diffusion currents adopt the exponential voltage dependence of the minority carrier densities p_n*, n_p*. Apart from these quantities, the currents depend on the diffusion constants and the diffusion lengths L_p, L_n which are determined by the respective minority carrier lifetime in the neutral regions.

Whereas the recombination in the neutral regions is considered to be essential, *the recombination/generation in the thin space charge layer is neglected* in the ideal I–V characteristic. Since $dj_{n,p}/dx = \pm q \cdot R$, this means that the electron and hole currents are assumed constant across the space charge layer. With $j_n(x_n) = j_n(x_p)$ one obtains

$$j = j_n(x_n) + j_p(x_n) = j_n(x_p) + j_p(x_n) \tag{3.49}$$

Thus, the current density j is given by the sum of the minority carrier diffusion currents in the neutral regions at the borders to the space charge layer, by adding (3.46) and (3.47), the current-voltage-characteristics of the pn-junction is obtained as

$$j = j_s \cdot \left(e^{\frac{q \cdot V}{k \cdot T}} - 1 \right) \tag{3.50}$$

with

$$j_s = q \cdot n_i^2 \cdot \left(\frac{D_p}{L_p \cdot N_D} + \frac{D_n}{L_n \cdot N_A} \right) \tag{3.51}$$

Equation (3.50) with (3.51) is the ideal current-voltage-characteristic of the pn-junction as derived by Shockley [Sho49]. In Fig. 3.10 the characteristic is shown in the normalized form j/j_s versus qV/kT. The current increases exponentially with positive voltage, in the blocking direction it approaches quickly the saturation current which is very small. If for example

$$\mathrm{N_A} = 1 \times 10^{16}\,\mathrm{cm^{-3}}, \mathrm{N_D} = 1 \times 10^{15}\,\mathrm{cm^{-3}},$$
$$\mathrm{L_n} = \mathrm{L_p} = 50\,\mu\mathrm{m}, \mathrm{D_n} = 30\,\mathrm{cm^2/s}, \mathrm{D_p} = 12\,\mathrm{cm^2/s}$$

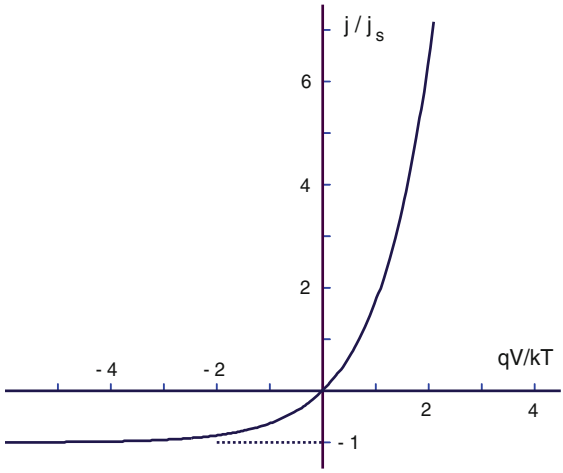


Fig. 3.10 Normalized ideal I-V characteristic of a pn-junction

one obtains for silicon with n_i (300 K) = 1.07 × 10^{10} cm^{-3}: j_s = 5.5 × 10^{-11} A/cm^2. The characteristic (3.50) in this general form has been derived first for the Cu/CuO_2 rectifier by Wagner [Wag31]. Also the later theory of the metal-semiconductor contact leads to the characteristic (3.50), the saturation current however differs essentially from (3.51). The exponential dependence of current on voltage is always based on the Boltzmann distribution.

In (3.51) the factor n_i^2 finally contains the dependence on the band gap and the main part of temperature dependence of j_s. In Fig. 3.11 the characteristics for several semiconductors at 300 K as given by (3.50), (3.51) are plotted on a larger scale than in Fig. 3.10, the reverse current is scaled logarithmically. For all semiconductors the same values of N_A, N_D and $D_{n,p}/L_{n,p}$ were used as in the above example for silicon. The saturation current density varies over many orders of magnitude. Up to a certain forward voltage, about 0.7 V for Si, the current remains very small on the scale of normal conduction current densities, whereas above that voltage it increases soon strongly. This threshold voltage can be defined with some arbitrariness as the voltage belonging to a current density j_{thr} of about 5 A/cm^2. Solving (3.50) for V

$$V = \frac{k \cdot T}{q} \ln\left(\frac{j}{j_s} + 1\right) \tag{3.52}$$

and inserting j = 5 A/cm^2 the threshold voltage V_{thr} in the case of Fig. 3.11 is obtained to be 0.26 V for Ge, 0.65 V for Si,[2] 1.09 V for GaAs and 2.77 V for 4H-SiC. These values are close to the built-in voltage V_{bi} calculated by Eq. (3.6a).

[2]The threshold voltage given in data sheets for power diodes is about 0.2–0.5 V higher. Because these diodes have a pin structure (see Sect. 5), a voltage drop across the base (i) region and at the i-n junction have to be added. As threshold voltage one uses there the onset voltage V_s in the simplified ohmic characteristic $V_F = V_s + R_{diff} I_F$.

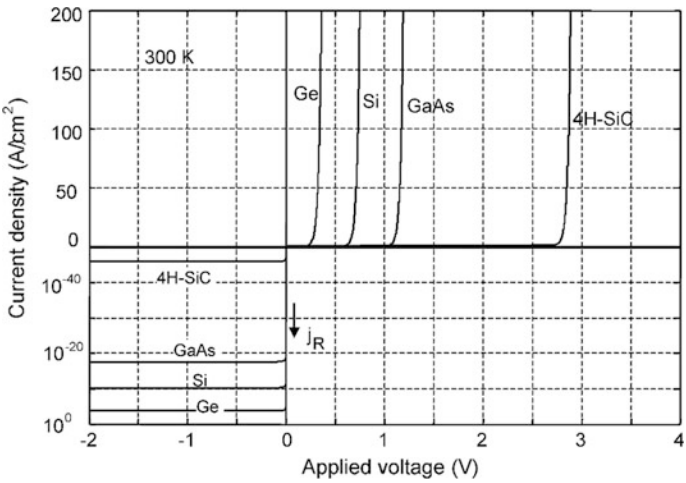


Fig. 3.11 Current density versus applied voltage for ideal pn-junctions in different semiconductors at 300 K

The threshold current, although near the lower end of normally used current densities, is already so high that the applied voltage must balance already largely the built-in voltage. The threshold voltage increases nearly linearly with band gap as follows from (3.52) if Eq. (2.6) is inserted. Experiments are well reproduced on the whole by this theory.

Of course a small threshold voltage is advantageous because of the forward losses. On the other hand, the high j_s required for small V_{thr} means a high reverse leakage current. In the case of Ge this disadvantage outweighs strongly the advantage of small threshold voltage. Already at 100 °C the leakage current grows so high that it results in a hardly controllable heating. For wide-gap semiconductors on the other hand, the large threshold voltage is the more weighing disadvantage. Here, the large threshold voltage of pn-junctions is often avoided by using a metal-semiconductor junction and for switching a unipolar field effect transistor without junction in the current path.

In practical devices, the pn-junctions are mostly very asymmetric. With $N_A \gg N_D$ Eq. (3.51) simplifies to

$$j_s = q \cdot n_i^2 \cdot \left(\frac{D_p}{L_p \cdot N_D} \right) \tag{3.53}$$

The saturation current is determined in these cases only by the minority carrier parameters of the weakly doped zone. Using this equation, we consider now the *temperature dependence* of the voltage V at a given forward current, which is often used to indicate and control the temperature in integrated power devices. Inserting Eq. (2.6) for n_i^2 one obtains from (3.53), (3.52) for $V > 3 \cdot kT/q$:

$$V(j,T) = \frac{E_g}{q} - \frac{kT}{q} \cdot \ln\left(\frac{q \cdot D_p \cdot N_C \cdot N_V}{L_p \cdot N_D \cdot j}\right) \tag{3.54}$$

Because the term $q \cdot D_p \cdot N_C \cdot N_V/(L_p \cdot N_D)$ represents a very high current density whose temperature dependence due to the logarithm has little influence, the forward voltage V decreases nearly linearly with increasing T. In Fig. 3.12, the dependency (3.54) with use of Eqs. (2.8) and (2.9) is plotted for two current densities for the example: $N_D = 1 \times 10^{16}$ cm^{-3} and $\tau_p(T) = 1\ \mu s \cdot (T/300)^2$. This temperature dependence is found often and leads to $D_p/L_p = \sqrt{(D_p/\tau_p)} \approx 3.3 \times 10^3\ (300/T)^{3/2}$ cm/s.

Regarding the conditions underlying the Shockley characteristic (3.50), (3.51), the assumption of negligible recombination in the space charge region is sometimes not satisfied in silicon even at forward bias where the thickness of the space charge layer is very small. This can be seen, if at small forward current densities, for example in the range of 100 js to 1 A/cm^2, the measured current is plotted on a logarithmic scale versus voltage. The observed slope is often not q/kT as required by (3.50) but essentially smaller. The measured characteristic can be described then by

$$j = j_s \cdot \left(e^{\frac{q \cdot V}{n \cdot k \cdot T}} - 1\right) \tag{3.55}$$

where the number n ranges often between 1.7–2. Because of the small current densities this cannot be attributed to a resistive voltage drop in the p and n region. The recombination in the space charge layer is given by the SRH Eq. (2.68) with $n \cdot p = const \gg n_i^2$ at forward bias [see Eq. (3.39)]. The nominator of this equation has approximately the same magnitude as in the neighboring parts of the neutral regions. However, if the recombination level lies near the middle of a not too small band gap and hence the concentrations n_r and p_r both are small, the denominator in (2.68) is on average small against the value in the neutral regions where n or p is large. Hence the recombination rate R is very high, and in spite of the small

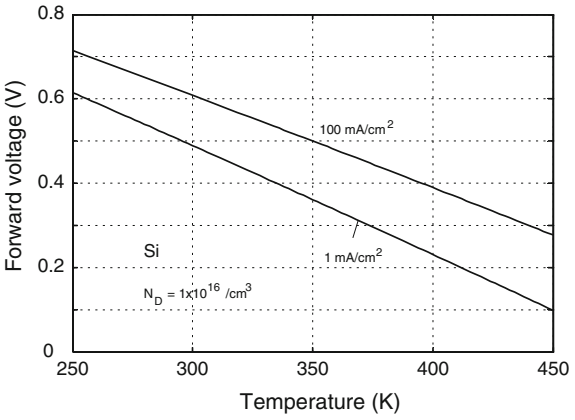


Fig. 3.12 Temperature dependence of forward voltage of a p$^+$n-junction at constant current densities

thickness of the space charge layer the integral recombination $\int Rdx$ can be significant. Only if the recombination level is located in considerable distance from the middle of the band gap, the recombination in the space charge layer is negligible, and the ideal characteristic is measured.

At current densities, where the resistive voltage drop V_{drift} caused by the majority carrier currents in the p and n region is considerable, the voltage V in the above equations has to be interpreted as the junction voltage $V_j = V_F - V_{drift}$ where V_F is the total forward voltage. As long as the resistance is given by the doping concentration and thicknesses of the two semiconductor regions and hence is a constant, V_F as function of j is obtained by adding the drift voltage $V_{drift} = r \cdot j$ (r = resistance times area) to the right hand side of Eq. (3.52):

$$V_F = \frac{kT}{q} \cdot \left(\ln \frac{j}{j_s} + 1 \right) + r \cdot j$$

A current independent resistance is tied however to the condition of low injection levels, which was assumed also for the derivation of Eq. (3.50). This condition is satisfied over a wide range of current densities only for relatively high doping concentrations of the p and n region. For power diodes with a p^+nn^+ structure, the resistance of the weakly doped base region is strongly modulated at usual forward current densities by high injection of carriers. The characteristics of power diodes are treated in Chap. 5.

3.3 Blocking Characteristics and Breakdown of the pn-Junction

3.3.1 Blocking Current

Especially for reverse voltages, the Eqs. (3.50), (3.51) agree with measurements only in limited ranges of voltage and temperature if at all. Often the measured blocking current is much higher than the saturation current (3.51) because of large generation in the space charge region. A second effect not included in (3.50), (3.51) is avalanche multiplication which at high reverse voltages leads to enhanced blocking current and at a certain voltage to complete breakdown of the blocking ability. This effect will be treated in the next section, in the present section we consider the blocking current as made up of the saturation current (3.51) and the current caused by generation in the space charge region.

Since a reverse voltage results in a negative excess carrier concentration (deviation from equilibrium), the thermal generation rate $G(x)$ is everywhere ≥ 0. The total reverse current density j_r owing to thermal generation is obtained by integrating the continuity equation $dj_p/dx = -q \cdot R = q \cdot G$ from the left border of the p region ($x = -\infty$) to the right border of the n region ($+\infty$). This yields:

$$q\int_{-\infty}^{\infty} G dx = j_p(\infty) - j_p(-\infty) = -j_p(-\infty) = -j = j_r, \tag{3.56}$$

since the hole current in the p region at large distance from the junction equals the total current. For the calculation, one uses as before the abrupt depletion approximation with $n \approx p \approx 0$ in the space charge region and with the minority carrier distribution in the neutral regions as given for the holes in the n region by (3.45), where now $V < 0$ (see Fig. 3.9c). The integral over the neutral regions delivers the reverse current of (3.50), (3.51), as can be easily verified: The minority carriers diffusing out of the neutral regions are *generated* there due to the negative excess concentration. For the current caused by generation in the *space charge region,* the formula (2.68) applies. In order that the terms with n and p can be neglected on the base of (3.39) and (3.15), (3.18), the reverse voltage V_r is assumed to be $>3 \cdot kT/q$. The generation rate is then constant and given by (2.77), and the reverse current generated in the space region (boundaries x_p and x_n, thickness $w = x_n - x_p$) is

$$j_{sc} = q \cdot \int_{x_p}^{x_n} G dx = \frac{q \cdot n_i \cdot w}{\tau_g} = \frac{q \cdot N_r \cdot w}{1/e_n + 1/e_p} \tag{3.57}$$

Since the charge density on both sides of the junction is given by the doping concentration in the used approximation and is therefore independent of the voltage, the extension of the space charge region into the p and n region (and other relationships) are given by the same expressions as for zero bias except that V_{bi} has to be replaced by $V_{sc} = V_{bi} - V = V_{bi} + V_r$. Hence Eq. (3.19a) for an abrupt p^{+-} n-junction generalizes to

$$w = \sqrt{\frac{2 \cdot \varepsilon \cdot (V_{bi} + V_r)}{q \cdot N_D}} \tag{3.58}$$

Inserting this and adding (3.57) to (3.53) the total blocking current density of a p^+n-junction is obtained as:

$$j_r = j_s + j_{sc} = q \cdot \left(\frac{n_i^2}{N_D} \cdot \frac{L_p}{\tau_p} + \frac{n_i}{\tau_g} \cdot \sqrt{\frac{2 \cdot \varepsilon \cdot (V_{bi} + V_r)}{q \cdot N_D}} \right) \tag{3.59}$$

where D_p/L_p is written as L_p/τ_p to express the two current contributions in a similar manner by a concentration, a length and a lifetime. Also for the diffusion current the condition $V_r > 3 \cdot kT/q$ is used to be in saturation.

Via the width of the depletion region the blocking current increases now with voltage. Whereas the diffusion current is proportional to n_i^2, the space charge term increases only linear with n_i. Hence the diffusion term in proportion to the space charge current increases with intrinsic concentration, that is with decreasing band

gap and increasing temperature. Which part predominates depends, however, also on the lifetime τ_g as compared to τ_p as well as on the voltage V_r. If the generation lifetime τ_g is comparable with τ_p and w comparable with L_p, the space charge term in (3.59) predominates to the extent as $n_i \gg p_{n0} = n_i^2/N_D$ or $n_i \ll N_D$. However, as has been discussed in Sect. 2.7.2, the generation lifetime depends exponentially on the recombination level and becomes very large if E_r is not located near the middle of the band gap. According to the Eqs. (2.70), (2.61) (2.63) and (2.69) of the Shockley-Read-Hall model, the generation lifetime in the depletion region can be written

$$\tau_g = \frac{n_i}{N_r}\left(\frac{1}{c_n \cdot n_r} + \frac{1}{c_p \cdot p_r}\right) = n_i \cdot \left(\frac{\tau_{n0}}{n_r} + \frac{\tau_{p0}}{p_r}\right) \tag{3.60}$$

whereas the low-level minority lifetime τ_p according to Eq. (2.72) is given by

$$\tau_p = \tau_{p0} + (\tau_{p0} \cdot n_r + \tau_{n0} \cdot p_r)/N_D \tag{3.61}$$

n_r and p_r are (except for the degeneracy factor) equal to the carrier concentrations obtained assuming the Fermi level equal to the recombination level [see (2.61a, (2.63a)]. If E_r coincides with the intrinsic level E_i, one has $n_r = p_r = n_i$ and (3.60), (3.61) yield $\tau_g = \tau_{n0} + \tau_{p0}$ which is close to the minimum of τ_g. On the other hand, if the recombination level is distinctly distant from the middle of the band gap, either n_r or p_r is small against n_i and hence $\tau_g \gg \tau_p$ according to (3.60). In this case the space charge current can be small compared with the saturation current.

In Fig. 3.13 the reverse current density of a p^+n-junction in Si is plotted versus the reciprocal absolute temperature for an acceptor level $E_a = E_i$ (intrinsic energy) as well as for a donor level $E_d = E_V + 0.32$ eV (both calculated with g = 2). The reverse voltage is 1000 V. The case of $E_a = E_i$ reflects the behavior of gold-doped junctions since gold has an acceptor level very near the middle of the band gap (see Fig. 2.19). The level 0.32 eV above the valance band coincides with the donor level of Pt. The two levels represent extreme cases regarding the reverse current. A level on the intrinsic energy is most unfavorable because the space charge current is maximal. As shown by the figure (dotted line) it accounts for the whole reverse current, except for the upper temperature range. This holds even at a small reverse voltage. A level 0.32 eV above E_V (or below the conduction band) on the other hand represents an optimal case: The reverse current is much reduced, caused by a very strong reduction of j_{sc} and to some extent also by a lower j_s resulting from the larger low-level lifetime τ_p in the neutral n region. The reverse current is now however nearly totally made up by the saturation current j_s, so it cannot be further reduced via j_{sc}. By a further increase of the distance of the level from the middle of the gap the reverse current can be lowered only relatively weak by a decrease of the saturation current j_s caused by a very high lifetime τ_p. Since this is connected with an extremely strong decrease of the lifetime with injection, it is not advantageous.

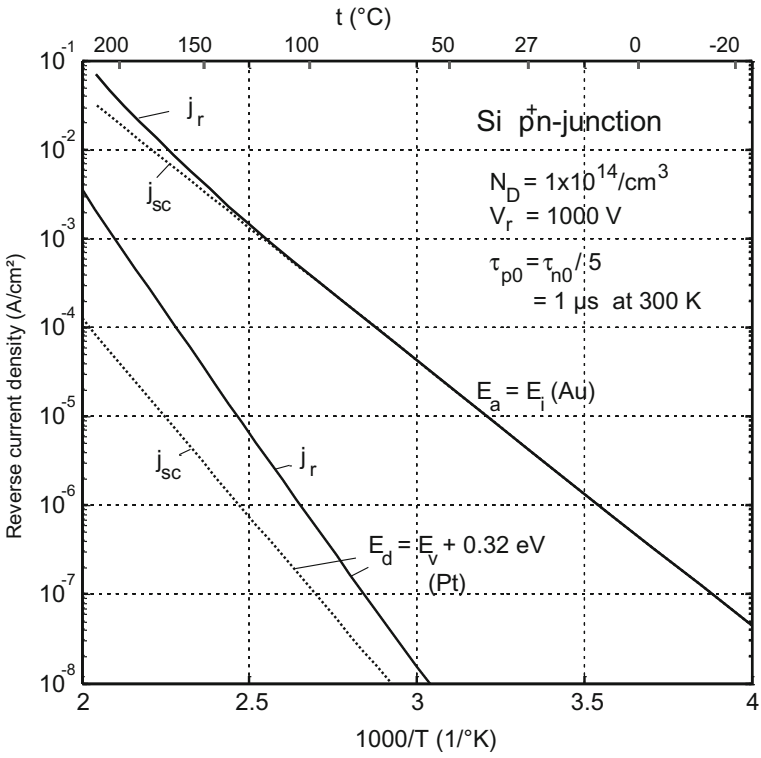


Fig. 3.13 Reverse current density of a p^+n-junction in Si as function of temperature for two recombination levels. In both cases the lifetime parameters τ_{n0}, τ_{p0} are assumed as $\tau_{p0} = \tau_{n0}/5 = 1\ \mu s \cdot (T/300)^2$. Degeneracy factor in both cases g = 2

Actually the data in Sect. 2.7.2 d yielded, that the *acceptor* level of Pt (0.23 eV below E_C) accounts for the main part of the space charge current of Pt (see page 80). Nevertheless the j_r-curve for the 0.32 eV level in Fig. 3.13 provides a useful approximate depiction of the reverse current behavior of platinum doped diodes.

The heat dissipation during reverse bias, $j_r \cdot V_r$, is usually much smaller than at forward conduction. However, because j_r increases strongly with temperature, the heat generation can increase stronger than the heat conduction to the sink. Since this can lead to an uncontrolled temperature increase (thermal run-away) and destruction, the reverse current density must be small enough. As an upper limit for diodes of the 1500-V class a value of about 10 mA/cm^2 at 150 °C is typically used. Such a limit implies that the recombination center density has to be limited, which in the case of gold has the consequence that otherwise possible switching times cannot be attained.

3.3.2 Avalanche Multiplication and Breakdown Voltage

At a certain breakdown voltage, the reverse current increases abruptly and the blocking ability is lost. Except for very small breakdown voltages, smaller than 10 V in Si, this is due to impact ionization or avalanche multiplication as described in Sect. 2.8. By the kinetic energy gained in the electric field a carrier raises an electron from the valence to the conduction band and creates an electron-hole pair, and these secondary particles again create electron hole pairs, thus an avalanche process is initiated. The effect is significant only at high field strengths occurring under high reverse bias. With the avalanche generation rate (2.89) the continuity equations (2.102) and (2.103) in the one-dimensional stationary case take the form

$$\begin{aligned} \frac{dj_p}{dx} &= \alpha_p \cdot j_p + \alpha_n \cdot j_n + q \cdot G \\ \frac{dj_n}{dx} &= -\alpha_p \cdot j_p - \alpha_n \cdot j_n - q \cdot G \end{aligned} \tag{3.62}$$

where α_n, α_p are the field-depend impact ionization rates and G as before denotes the thermal generation rate. The total current density

$$j_n + j_p = j \tag{3.63}$$

is independent of x under stationary conditions (see Eq. (2.106)). The avalanche effect is expressed in (3.62) by the proportionality of the generation rate to the current densities which at the relevant field strengths are proportional to the carrier densities. The course of the field strength, of the ionization rates and current densities is shown for a p^+n-junction in Fig. 3.14 (see the legend). To have positive current densities and field strength at reverse bias, the n region is at the left and the p region at the right hand side. The strong variation of j_n, j_p in the high field region due to avalanche multiplication is illustrated qualitatively.

For a qualitative understanding we assume at first that the ionization rates are equal, $\alpha_n = \alpha_p \equiv \alpha$, as is found e.g. in GaAs. Then Eqs. (3.62) with (3.63) simplify to

$$\frac{dj_p}{dx} = \alpha \cdot j + q \cdot G = -\frac{dj_n}{dx} \tag{3.64}$$

Integrating over the space charge region with the boundaries $x_n = 0$ and $x_p = w$ one obtains

$$j - j_{ns} - j_{ps} = j \cdot \int_0^w \alpha \cdot dx + q \cdot w \cdot G$$

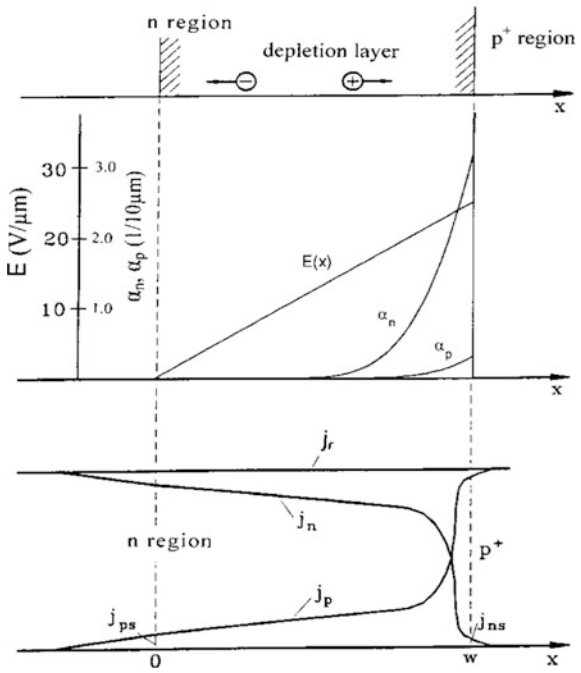


Fig. 3.14 Reverse biased p^+n-junction with avalanche multiplication: Field strength, ionization rates and reverse current densities as function of x. The ionization rates in dependency on the field strength refer to Si

since $j_p(x_p) = j - j_{ns}$ and $j_p(x_n) = j_{ps}$, where j_{ns}, j_{ps} are the minority carrier saturation current densities entering the depletion layer from the neutral regions. The current density j follows as

$$j = \frac{j_{ns} + j_{ps} + j_{sc}}{1 - \int_0^w \alpha \, dx} \tag{3.65}$$

where the previous notation $j_{sc} = q \cdot w \cdot G$ is used. As shown by this equation, the effect of avalanche can be expressed by a multiplication factor

$$M = \frac{1}{1 - \int_0^w \alpha \, dx} \tag{3.66}$$

which enhances the 3 components of the thermal current density. If with increasing voltage the field via the ionization rate satisfies the equation

$$\int_0^w \alpha(\mathsf{E}(x))dx = 1 \tag{3.67}$$

the current increases to infinity. Hence this is the condition from which the breakdown voltage $V_r = V_B = \int_0^w \mathsf{E} \, dx$ can be calculated. Below the breakdown voltage, Eq. (3.65) describes the enhancement of the reverse current by avalanche multiplication.

The considered case $\alpha_n = \alpha_p$ allows a simple calculation because the position where an impact ionization initiates the avalanche process has no influence on the total generated avalanche charge in this case: After generation of an electron-hole pair at a point x the electron moves to the neutral n region and the hole to the p region, so both together cover the whole width of the space charge region (high field region). This holds also for the pairs generated by the secondary carriers at a point x'. Hence if electrons and holes are equally effective, the total generated avalanche charge is independent of x. This is also the cause why all three current components j_{ns}, j_{ps} and j_{sp} are multiplied by the same factor.

If $\alpha_n \neq \alpha_p$ as in the case of Si and most other semiconductors (see Figs. 2.23 and 2.24), the calculation is much more complicated. In Appendix C it is shown that the avalanche multiplication can be expressed generally by double integrals over the ionization rates. As expected, the three components of the thermal blocking current (nominator of (3.65)) are enhanced for $\alpha_n \neq \alpha_p$ each by a different multiplication factor and hence the leakage current density is generally given by

$$j = M_n \cdot j_{ns} + M_{SC} \cdot j_{SC} + M_p j_{ps} \tag{3.68}$$

From $\alpha_n > \alpha_p$ in the case of Si, it follows that $M_n > M_{SC} > M_p$. All three M's tend to infinity at the same voltage, the breakdown voltage. For an abrupt p^+n-junction in Si the multiplication factors, as calculated numerically from the formulae in the appendix using the ionization rates of Eqs. (2.96), are plotted in Fig. 3.15 versus the reverse voltage. The breakdown voltage for the assumed doping density 1×10^{14} cm^{-3} of the n region is 1640 V. Already at 1400 V, the current density j_{ns} is enhanced by $M_n = 5$, whereas the multiplication factor M_p amounts only to 1.17 at this voltage. In the case of a single p^+n-junction the higher M_n is not significant for the blocking current, since the electron saturation current j_{ns} is very small

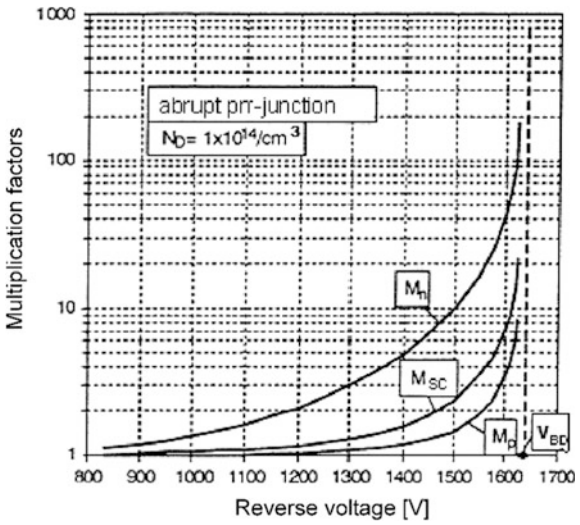


Fig. 3.15 Multiplication factors in dependence of the voltage for an abrupt p^+n-junction with $N_D = 1 \times 10^{14}$ cm^{-3}, $T = 300$ K

according to (3.51) due to the high N_A. For a transistor structure with a p^+n-junction between base and collector, however, the large M_n has a strong effect on the breakdown voltage, as will be seen.

Analytically, the multiplication factors as function of voltage are approximated often by the equation [Mil57]

$$M = \frac{1}{1 - (V/V_B)^m} \tag{3.69}$$

where the exponent m is used for fitting. To approximate the dependencies of Fig. 3.15, very different m-values are necessary for M_n and M_p. Fitting in both cases at $M_{n,p} = 2$, a value $m = 2.2$ is obtained for M_n and $m = 13.2$ for M_p. These values differ significantly from values in the literature where often m = 4 and 6 are used for M_n, M_p respectively. Although (3.69) is only a rough approximation, considering the whole voltage range, it is very useful for dimensioning of transistor and thyristor structures. The large difference between M_n and M_p has a significant influence on the blocking behavior.

If only the breakdown voltage of a (single) pn-junction is of interest, the complicated ionization integrals in the Appendix C can be avoided. As noted already in Sect. 2.8 and shown in Appendix C, the breakdown voltage can be calculated using the effective ionization rate

$$\alpha_{eff} = \frac{\alpha_n - \alpha_p}{\ln(\alpha_n/\alpha_p)} \tag{3.70}$$

which at breakdown satisfies the condition [Wul60, Oga65]

$$\int_0^w \alpha_{eff}(\mathsf{E}(x))dx = 1 \tag{3.71}$$

similarly as α in (3.67). This holds exactly only, if the ratio α_n/α_p is independent of E. However, also if this is not well fulfilled over a large field range, (3.71) is often a good approximation, because only a relative small field range contributes significantly to the integral.

The maximum field strength in the depletion layer, at which breakdown occurs, is called the critical field strength E_c. If the width of the high-field region is varied via the doping density according to (3.58) for a p^+n-junction, this must be compensated according to (3.71) by an inverse variation of α_{eff}. However, due to the very strong increase of the α's with E, this requires only a small change of the field strength. Hence E_c depends only weakly on the thickness w of the depletion region, and for rough estimates E_c is assumed often as constant.

Without any restriction we assume at first that the critical field (in dependence of w and N_D) is known and ask how the width and doping concentration of the weakly doped region(s) of a pn-junction are to be chosen to reach a desired breakdown

voltage V_B. Considering an abrupt p^+n-junction as shown in Fig. 3.14 the breakdown voltage is

$$V_B = \int_0^w \mathsf{E} dx = \frac{1}{2} w \mathsf{E}_c \tag{3.72}$$

Within the approximation of constant E_c the breakdown voltage is proportional to the width w of the space charge region. According to the Poisson equation

$$\frac{d\mathsf{E}}{dx} = \frac{\mathsf{E}_c}{w} = \frac{q \cdot N_D}{\varepsilon} \tag{3.73}$$

w depends inversely on the doping density N_D. Expressing the width w in (3.72) by N_D one obtains

$$V_B = \frac{\varepsilon \cdot \mathsf{E}_c^2}{2 \cdot q \cdot N_D} \tag{3.74}$$

Hence, within the approximation of constant E_c, the breakdown voltage is inversely proportional to the doping density of the n region. However, as mentioned, this is only a very rough approximation.

To calculate the critical field in dependence of w or N_D in analytical form, we use the power approach (2.94) in the form [Shi59, Ful67]

$$\alpha_{eff} = B \cdot \mathsf{E}^n \tag{3.75}$$

With the coordinates of Fig. 3.14, the field strength is

$$\mathsf{E}(x) = \frac{q\, N_D}{\varepsilon} x \tag{3.76}$$

Inserting (3.75) together with $dx = \varepsilon/(q \cdot N_D) \cdot d\mathsf{E} = w/\mathsf{E}_c \cdot dE$ into the condition (3.71) one gets

$$\frac{B\, w}{\mathsf{E}_c} \int_0^{\mathsf{E}_c} \mathsf{E}^n \, d\mathsf{E} = B\, w \frac{\mathsf{E}_c^n}{n+1} = 1 \tag{3.77}$$

$$\mathsf{E}_c = \left(\frac{n+1}{B \cdot w} \right)^{\frac{1}{n}} = \left(\frac{q \cdot (n+1) \cdot N_D}{B \cdot \varepsilon} \right)^{\frac{1}{n+1}} \tag{3.78}$$

The last expression follows from (3.77) with $w = \varepsilon \cdot \mathsf{E}_c/(q \cdot N_D)$ [Eq. (3.73)]. As expected, E_c decreases slightly with increasing w and increases slightly with N_D.

Inserting (3.78) into (3.74) the breakdown voltage as function of doping density is obtained as

$$\begin{aligned} V_B &= \frac{\varepsilon}{2 \cdot q \cdot N_D} \cdot \left(\frac{q \cdot (n+1) \cdot N_D}{B \cdot \varepsilon} \right)^{\frac{2}{n+1}} \\ &= \frac{1}{2} \cdot \left(\frac{n+1}{B} \right)^{\frac{2}{n+1}} \cdot \left(\frac{\varepsilon}{q \cdot N_D} \right)^{\frac{n-1}{n+1}} \end{aligned} \tag{3.79}$$

(3.72) and (3.78) yield

$$V_B = \frac{1}{2} \cdot \left(\frac{n+1}{B} \cdot w^{n-1} \right)^{\frac{1}{n}} \tag{3.80}$$

The width w of the depletion region as function of N_D at breakdown follows from (3.78) as

$$w = \left(\frac{n+1}{B} \right)^{\frac{1}{n+1}} \cdot \left(\frac{\varepsilon}{q \cdot N_D} \right)^{\frac{n}{n+1}} \tag{3.81}$$

Due to the variation of the critical field, the increase of V_B with w and with $1/N_D$ is sub-linear.

To adjust (3.75) to the Chynoweth law, the constants n and B are to be chosen according to (2.95), (2.94). With the numerical Eq. (2.92) for α_{eff} in Si at 300 K, one obtains in dependence of the field E_0 where the adjustment is carried out:

$$n = \frac{1.68 \times 10^6 \text{ V/cm}}{\mathsf{E}_0} \tag{3.82}$$

$$B = \frac{1.06 \times 10^6 \text{ V/cm}}{\mathsf{E}_0^n \cdot \exp(n)} \tag{3.83}$$

Following Shields [Shi59], most authors set the exponent to $n = 7$ in silicon. According to (3.82) this is obtained at $\mathsf{E}_0 = 2.4 \times 10^5$ V/cm. The corresponding B-value according to (3.83) is $B = 2.107 \times 10^{-35}$ cm^6/V^7. Using these constants, Eqs. (3.80), (3.81) can be written

$$V_B = \frac{1}{2} \cdot \left(\frac{8}{B} \right)^{\frac{1}{4}} \cdot \left(\frac{\varepsilon}{q \cdot N_D} \right)^{\frac{3}{4}} = 563 \text{ V} \cdot \left(\frac{4 \times 10^{14}/cm^3}{N_D} \right)^{\frac{3}{4}} \tag{3.84}$$

$$w = \left(\frac{8}{B} \right)^{\frac{1}{8}} \cdot \left(\frac{\varepsilon}{q \cdot N_D} \right)^{\frac{7}{8}} = 42.6\,\mu\text{m} \cdot \left(\frac{4 \times 10^{14}/\text{cm}^3}{N_D} \right)^{\frac{7}{8}} \tag{3.85}$$

In the numerical expressions on the right hand side, the doping density is related to the value 4×10^{14} cm^{-3} which is roughly the point at which the approximation is adapted and around which it will be very accurate. It is taken into account that the integral of α_{eff} (E) over the triangular field distribution (see Fig. 3.14) is best approximated choosing E_0 near $0.9 \cdot E_c$. Hence the choice $n = 7$, $E_0 = 2.4 \times 10^5$ V/cm means that the maximum field at the matching point is $E_c = 2.4 \times 10^5/0.9 = 2.667 \times 10^5$ V/cm. This field corresponds to the doping density $N_D = 4.36 \times 10^{14}$ cm^{-3} according to (3.78).

At much higher and lower voltages the accuracy suffers from the deviation of the power approximation from the Chynoweth law as illustrated in Fig. 2.23. In Fig. 3.16, the exponent n and the constant B are plotted versus the field E_0 where the fitting is carried out. n varies nearly by a factor 3 in the relevant field range. The extreme increase of B with E_0 is caused according to (3.83) by the decrease of n and the resulting rapid decrease of E_0^n. To take into account these variations, it is suitable to choose the matching point E_0 for each doping density separately as $0.9 \cdot E_c$ (N_D).

Inserting (3.82), (3.83) with $E_0 = 0.9 \cdot E_c$ into the right hand side of (3.78), one can solve the equation for E_c by iteration. With the critical field the breakdown voltage and width of the space charge layer are given by (3.74), (3.73) or can be calculated from (3.79), (3.81), since with E_c one has E_0 and then n and B. The breakdown voltage and width w at breakdown obtained in this way are plotted in Fig. 3.17 as functions of N_D. These results are very close to those calculated from the exact ionization integral given in Appendix C. In the figure also the $N_D^{-3/4}$ dependency according to Eq. (3.84) is shown (dotted line). As is seen, this simple formula is a good approximation over a wide range. For doping densities $N_D < 10^{14}$ and $>10^{15}$ cm^{-3} it underestimates the breakdown voltage. The used approach allows an analytical integration of (3.71) also in other cases where the field strength depends linearly on x. For $p^+n^-n^+$-diodes this is utilized in Chap. 5.

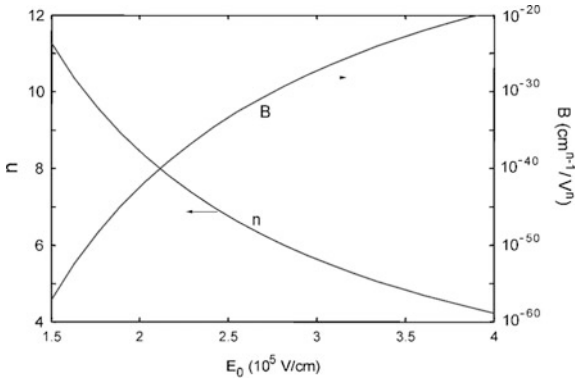


Fig. 3.16 Constants in the power law $\alpha_{eff} = B \cdot E^n$ as functions of the field strength at which it is fitted to $\alpha_{eff} = a/\exp(b/E)$. $a = 1.106 \times 10^6$ /cm, $b = 1.68$ V/cm (Si at 300 K)

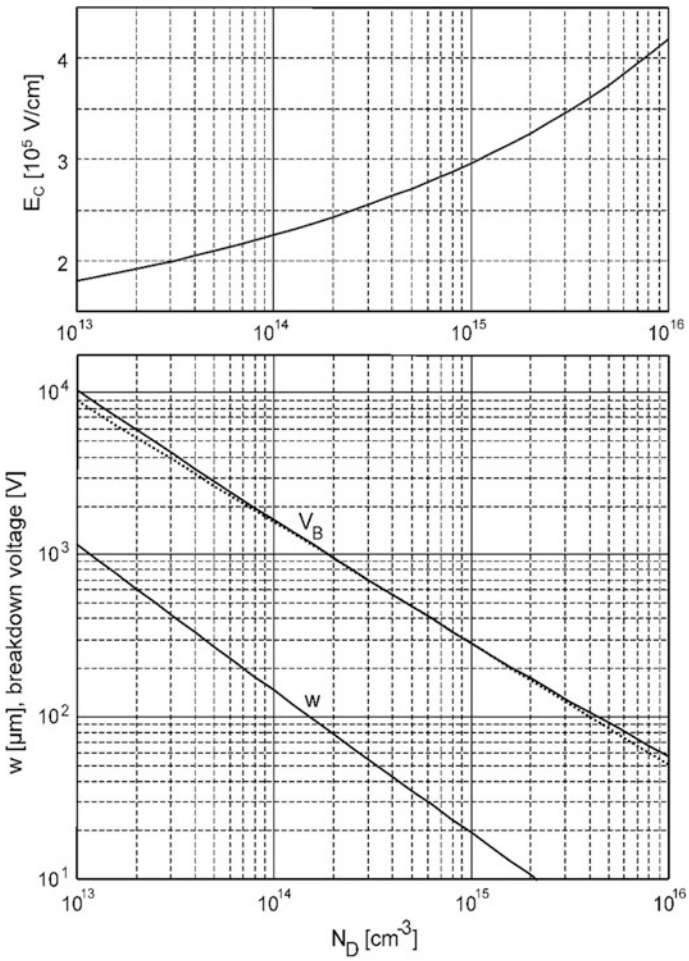


Fig. 3.17 Critical field strength, breakdown voltage and depletion width at breakdown as functions of the doping density N_D for abrupt p^+n-junctions in Si at 300 K. The dotted straight line represents the approximation (3.84)

Since often used, the explicit dependencies of N_D, w_B and E_c on the breakdown voltage are still noted. The inversion of (3.79) and (3.80) yields:

$$N_D = \frac{\varepsilon}{q}\left(\frac{n+1}{B}\right)^{\frac{2}{n-1}}\left(\frac{1}{2V_B}\right)^{\frac{n+1}{n-1}} \tag{3.86}$$

$$w_B = \left(\frac{B}{n+1}\right)^{\frac{1}{n-1}}(2V_B)^{\frac{n}{n-1}} \tag{3.87}$$

From Eq. (3.77), right hand part, one obtains inserting $w_B = 2V_B/\mathsf{E}_c$:

$$\mathsf{E}_c = \left(\frac{n+1}{2BV_B}\right)^{\frac{1}{n-1}} \tag{3.88}$$

These equations will be used later at several points in the book.

For diffused junctions with a very steep profile, the results for abrupt junctions can be used as an approximation. Generally, the approach (3.75) is useless for diffused junctions, because the integration of (3.71) or the ionization integral can be carried out even with this approximation only numerically. For a given doping density of the lightly doped region diffused junctions have a higher breakdown voltage than abrupt ones. This follows because (i) the maximum field at the junction is reduced and (ii) the field extends considerably into the highly doped diffused region near the junction. In Fig. 3.18 the numerically calculated breakdown voltage of diffused p^+n-junctions is plotted versus the background doping density N_D for two examples of steepness of the diffusion profile. The acceptor density around the junction at $x = 0$ is approximated here by an exponential decrease $N_A(x) \sim \exp(-x/\lambda)$ (orientation as in Fig. 3.6). Hence, λ is the decay length of the diffusion profile near the junction. Since $N_A(x) = N_D$ at $x = 0$, the net doping density in the space charge region is

$$N(x) = N_D - N_A(x) = N_D \cdot (1 - e^{-x/\lambda}) \tag{3.89}$$

and the doping gradient at the junction is $dN/dx(0) = -N_D/\lambda$. The case $\lambda = 3$ µm is typical for pn^-p structures in thyristors diffused with gallium or boron, the case $\lambda = 10$ µm is representative for high voltage pnp-structures realized with very deep aluminum diffusion.

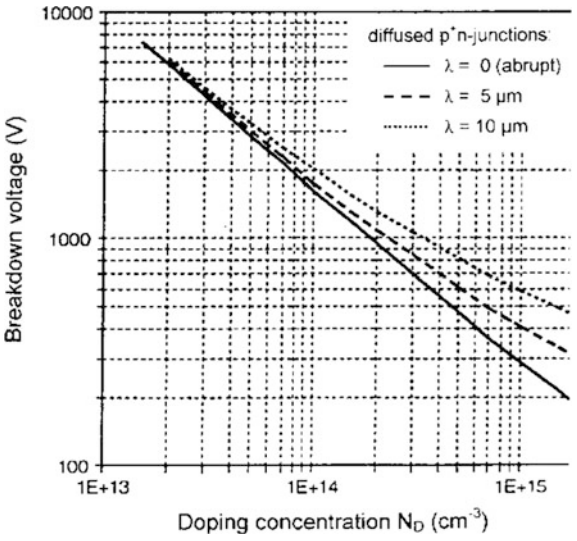


Fig. 3.18 Breakdown voltage of diffused p^+n-junctions in Si. λ is the decay lengths of the diffusion profile in the depletion layer

Temperature dependence

As described in Sect. 2.8, the ionization rates decrease with increasing temperature; for the effective ionization rate this is expressed numerically by (2.97). Hence the critical field strength and avalanche breakdown voltage increase slightly with T. Using again the power approach (3.75) to give an analytical description, the exponent n and constant B are needed as function of temperature. This is obtained from Eq. (2.98) together with (3.75):

$$\begin{aligned} n(T) &= \frac{b_{eff}(T)}{\mathsf{E}_0} = \frac{1.68 \times 10^6 + 1100 \cdot (T - 300)}{\mathsf{E}_0} \\ B(T) &= \frac{C(T)}{\mathsf{E}_0^{n(T)}} = \frac{a_{eff}}{\mathsf{E}_0^{n(T)} \exp(n(T))} = \frac{1.06 \times 10^6}{\mathsf{E}_0^{n(T)} \cdot \exp(n(T))} \end{aligned} \tag{3.90}$$

where E_0 is scaled in V/cm. Inserting these dependences in Eq. (3.79), the breakdown voltage for a triangular field distribution is given analytically as function of temperature. If for example the doping concentration is $N_D = 1 \times 10^{14}$ cm^{-3}, a good choice for E_0 is 2.2×10^5 V/cm (see Fig. 3.17). Then at 300 K Eq. (3.90) yields $n = 7.64$, $B = 7.78 \times 10^{-39}$ cm^{n-1}/V^n. With these values Eq. (3.79) gives $V_B = 1630$ V. At $T = 400$ K one obtains $n = 8.14$, $B = 1.01 \times 10^{-41}$ cm^{n-1}/V^n, and (3.79) results in $V_B = 1835$ V, a 12.6% higher value than at 300 K. Due to a considerable decrease of V_B from room temperature downwards, one has to dimension so that the targeted blocking ability is maintained still at the lower limit of operation temperature ($\approx$250 K).[3] This holds for diodes. For thyristors, the temperature dependency is weaker (and can be inverted), because the blocking voltage is supplied here by a transistor structure and depends not only on avalanche multiplication but also on the current amplification factor of a transistor, which increases with temperature.

3.3.3 Blocking Capability with Wide-Bandgap Semiconductors

The presented theory is of course applicable also to other semiconductors, particularly to those which have a wider band gap than Si. Owing to the higher energy required to carry an electron from the valence into the conduction band, the critical field strength increases with band gap. A summary of raw values of the critical field is given in Table 3.1. Using these values in (3.72), (3.74) a rough estimate can be obtained about the needed width w of the depletion region and the allowed maximum doping density for a given blocking ability.

[3]Some manufacturers data sheets specify V_B at 25 °C, and for application at lower temperatures one must be aware of said decrease of V_B with temperature.

Table 3.1 Raw values of the critical field strength for Si and semiconductors with a wider band gap

	E_C[V/cm]
Si	2×10^5
GaAs	4×10^5
4H-SiC	3×10^6
GaN	$>3 \times 10^6$
C (diamond)	$1–2 \times 10^7$

For 4H-SiC even an E_c-value of 3.3 MV/cm was reported recently for local avalanche [Rup14]. Using the value given in the table for SiC a p^+n structure blocking 10 kV requires a base width $w = 2V_B/E_c = 67$ µm and allows a base doping density $N_D = \varepsilon E_c^2/(2qV_b) = 2.40 \times 10^{15}$ cm^{-3}. For a more accurate calculation the decrease of the critical field strength with thickness of the depletion region or decreasing doping density is to be considered. The parameters for the effective ionization rate in (3.75) are reported for 4H-SiC as $n = 8.03$ and $B = 2.18 \times 10^{-48}$ cm$^{7.03}$V$^{-8.03}$ [Bar09]. Inserting these into (3.78) the critical field strength is obtained as

$$E_c = 2.58 \cdot \left(\frac{N_D}{10^{16}/\mathrm{cm}^3}\right)^{0.111} \mathrm{MV/cm} \tag{3.91}$$

For the blocking capability Eq. (3.79) gives:

$$V_B = 1770 \cdot \left(\frac{10^{16}/\mathrm{cm}^3}{N_D}\right)^{0.78} \mathrm{V} \tag{3.92}$$

The breakdown voltage calculated in this way for an asymmetrical junction in 4H-SiC is shown in Fig. 3.19 together with that of silicon. As can be seen, the maximum doping concentration for a given blocking voltage, for example 1000 V, is for SiC two decades higher than for Si, a consequence of the critical field strength being an order of magnitude higher. Owing to the much higher doping concentration and the smaller possible base width, SiC-devices show superior characteristics regarding conduction and switching losses, as will be shown in detail later.

From measurements at different temperatures with 4H-SiC $p^+n^-n^+$ diodes Bartsch et al. [Bar09] obtained the following temperature dependence of the parameters n and B

$$n = 6.78 + 1.25 \cdot \frac{T}{300\mathrm{K}} \quad B = 3 \cdot 10^{-40} \cdot \exp\left(-18.74 \cdot \frac{T}{300\mathrm{K}}\right) \tag{3.93}$$

At $T = 300$ K this leads to (3.91) and (3.92). Also mature SiC merged pin-Schottky diodes show a temperature dependence of blocking voltage which is in good agreement with these results; this is displayed later in Sect. 6.5 with Fig. 6.15.

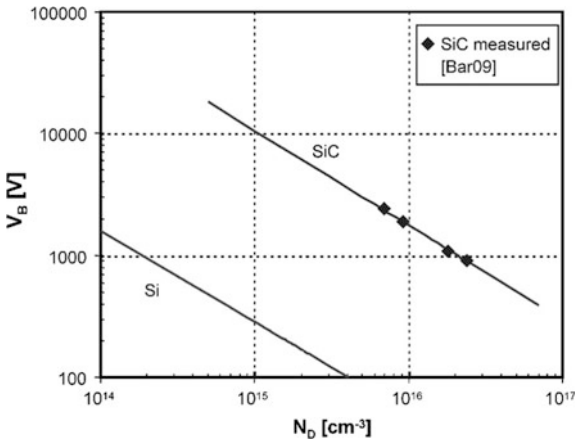


Fig. 3.19 Breakdown voltage of asymmetrical junctions in 4H-SiC and Si as function of the doping concentration of the weakly doped region

Several years it was practice in the design of SiC-devices to limit the electric field to a maximum value of about 1.5 MV/cm. The devices were specified for a lower blocking voltage than the breakdown voltage resulting from critical field strength, because the leakage current otherwise surmounts the allowed range. The reasons for the enhanced reverse current are crystal defects. In recent time, the SiC crystal quality has been improved strongly and former limits are overcome.

3.4 Injection Efficiency of Emitter Regions

In preceding sections mainly asymmetric pn-junctions have been considered. It turned out that to the extent that the doping ratio N_A/N_D is very large or small the influence of the properties of the highly doped region on the characteristics vanishes, in case of blocking as well as in forward direction. Since the density of injected minority carriers at forward bias is inversely proportional to the doping density according to (3.36), (3.37), mainly the highly doped region injects carriers into the weakly doped region and not vice versa. Because this is similar to the function of the emitter and base regions in transistors, the highly and weakly doped regions of asymmetrical junctions are also called emitter and base region, respectively. The injection efficiency, defined as the ratio of minority carrier current in the base region to the total current, is called emitter efficiency. A quantitative comparison of theory and experiment shows now that the measured emitter efficiency is essentially smaller than calculated on the base of the equations in Sect. 3.2. This manifests itself immediately in the current amplification factor of transistors. For power diodes with a p^+nn^+-structure, the current injected from the base into the emitter regions becomes much earlier significant with increasing injection level than according to the given 'classical' equations. A consequence is that the forward voltage drop of diodes and other power devices is significantly enhanced compared

with theoretical characteristics developed on the basis of above equations. The cause is that the band gap reduction in highly doped region(s) has been neglected till now.

In the present section we examine the injection properties of emitter regions with consideration of band gap narrowing. Furthermore, the injection level in the base region is allowed to be arbitrary and the high Auger recombination in the emitter is considered. According to the continuity Eq. (3.40), the minority carrier current entering the emitter is absorbed by recombination in the volume of the emitter and, if the thickness of the region is smaller than a few minority carrier diffusion length, at the contact. Although the contact recombination is utilized in novel devices to improve switching properties, we assume at first that only recombination in the volume is significant.

a) Characterization of emitter recombination:
The injection properties of a forward biased abrupt n^+p junction are examined first for the case of *arbitrary injection level in the base*, without considering already bandgap narrowing. The injection level in the emitter region is assumed to be small (see Fig. 3.20). According to Eqs. (3.3), (3.4), the ratio of carrier densities on one side to that on the other side of the space charge region are given by the Boltzmann factor containing the total potential difference between both sides:

$$\frac{p_n^*}{p_B^*} = \frac{n_B^*}{n^+} = e^{-q\cdot\Delta V/kT} \tag{3.94}$$

As illustrated in Fig. 3.20, the star indicates the carrier concentrations in the neutral regions at the boundaries to the space charge layer. The concentration n^+ in the emitter is equal to the donor density N_D due to the assumed low injection level there. For the minority carrier density in the emitter region (3.94) gives

$$p_n^* = \frac{n_B^* \cdot p_B^*}{n^+} = \frac{n_B^* \cdot (N_B + n_B^*)}{n^+} \tag{3.95}$$

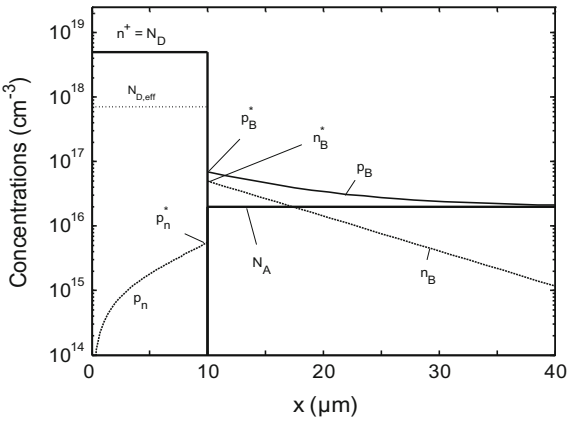


Fig. 3.20 Forward biased n^+p-junction with intermediate injection level in the p-base region

where the equation $p_B^* = N_B + n_B^*$ follows from neutrality. The base doping N_A is replaced here by N_B to include the case of an n^+n-junction, occurring in power diodes, by defining $N_B = -N_D$ in case of an n-base. For power diodes which have a low base doping, (3.95) is mainly significant for high injection levels in the base region where $p_B^* \approx n_B^*$, in bipolar transistors and thyristors the injection level in the p-base region is typically intermediate in the interesting current range. As shown earlier (see Eqs. (3.46) and (3.43)) the injected holes result in the minority current density in the n^+-region

$$j_p(x_n) = q \cdot \frac{D_p}{L_p} \cdot (p_n^* - p_{n0}) \tag{3.96}$$

Neglecting the very small equilibrium concentration $p_{n0} = n_i^2/N_D$ this can be written using (3.95)

$$j_p(x_n) = q \cdot h_n \cdot n_B^* \cdot p_B^* \tag{3.97}$$

where the properties of the n emitter are comprised in the constant:

$$h_n = \frac{D_p}{N_D \cdot L_p} = \frac{1}{N_D} \cdot \sqrt{\frac{D_p}{\tau_p}} \tag{3.98}$$

This parameter characterizes the emitter recombination which reduces the injection efficiency. It equals the saturation current density j_{ps} of the emitter defined in Eq. (3.46a) except that the factor $q \cdot n_i^2$ is omitted. It is useful to introduce the h-quantity, because the intrinsic concentration with its strong temperature dependence is not relevant for the injection efficiency for a given current. The h–parameter is nearly temperature-independent. The Eq. (3.97) defining the h-parameter by the proportionality between $j_p(x_n)$ and the product $n_B^* \cdot p_B^*$ is usable also for diffused emitter regions. The experimental h-values determined from (3.97) [Sco69, Bur75, Sco79] are not in close accordance with the saturation current density determining the I-V-characteristics at low current densities.

For the emitter efficiency of the n^+p-junction

$$\gamma = j_n(x_p)/j \tag{3.99}$$

one obtains with $j_n(x_p) = j - j_p(x_p) \approx j - j_p(x_n)$ by insertion of (3.97)

$$\gamma = 1 - j_p(x_n)/j = 1 - q \cdot h_n \cdot n_B^* \cdot p_B^*/j \tag{3.100}$$

As is seen, a small h_n–value is required to get a high emitter efficiency and current amplification.

According to (3.98) h_n can be made small by choosing a high doping density N_D. The explicit N_D dependence is however counteracted by the decrease of the minority carrier lifetime due to Auger recombination described in Sect. 2.7.1. For

$N_D > 5 \times 10^{18}$ cm^{-3} the lifetime is often solely determined by this recombination process, hence according to (2.56) one has $1/\tau_p = c_{A,n} \cdot N_D^2$. Inserting this, (3.98) yields the following "limiting" h-value for high N_D

$$h_{n,\mathrm{lim}} = \sqrt{D_p \cdot c_{A,n}} \tag{3.101}$$

which decreases however still slightly with increasing n$^+$ owing to the decrease of the diffusion constant D_p. Using the Auger coefficient of (2.57) and $D_p = 1.9$ cm^2/s for a doping concentration of 10^{19} cm^{-3}, (3.101) gives $h_{n,lim} = 7.3 \times 10^{-16}$ cm^4/s. Experimentally, the lowest h-values are an order of magnitude higher [Sco69, Bur75, Sco79, Coo83]. This is attributed to the band gap narrowing in the emitter region neglected till now.

b) Influence of bandgap narrowing:
In *thermal equilibrium* the ratio of electron concentrations in the weakly doped p region to that in the highly doped n$^+$ region is

$$\frac{n_{p0}}{N_D} = \frac{n_{i0}^2}{N_A N_D}$$

where n_{i0} is the intrinsic concentration in the p region (with low doping). The ratio of *hole* concentrations in the highly doped n$^+$ region to that in the p region is

$$\frac{p_{n0}}{N_A} = \frac{n_i^2}{N_A N_D} = \frac{n_{i0}^2 e^{\Delta E_g/kT}}{N_A N_D}$$

See Eq. 2.27 due to the bandgap narrowing the change of the hole concentration over the space charge region is enhanced compared with the change of electron concentration in the opposite direction:

$$\frac{p_{n0}}{N_A} = \frac{n_{p0}}{N_D} e^{\Delta E_g/kT} \tag{3.102}$$

If now a forward voltage V is applied, the associated potential change in the space charge region acts in the same manner on the holes as on the electrons, apart from the opposite direction of the force. Both sides of (3.102) are increased hence by the same factor $\exp(qV/kT)$. This is quite independent of the injection level, which in the p region is arbitrary. Hence with the notation of Fig. 3.20 it follows:

$$\frac{p_n^*}{p_B} = \frac{n_B^*}{N_D} e^{\Delta E_g/kT} \tag{3.103}$$

$$p_n^* = \frac{n_B^* p_B^*}{N_D} e^{\Delta E_g/kT} \tag{3.104}$$

where $p_B^* = N_A + n_B^*$. Defining an effective doping concentration $N_{D,eff} = N_D \cdot \exp(-\Delta E_g/kT)$, one has for the concentration ratios at opposite sides of the space charge layer for all doping densities $p_n^*/p_B^* = n_B^*/N_{D,eff}$ (see Fig. 3.20). This effective doping concentration increases only very little with increasing N_D. It serves only to illustrate the effect of band gap narrowing on the injection properties and has no significance at all for the majority carrier concentration $n^+ = N_D$. Inserting (3.104) into (3.96) the h_n quantity as defined by (3.97) takes the form

$$h_n = \frac{1}{N_D} \frac{D_p(N_D)}{L_p(N_D)} exp\left(\frac{\Delta E_g}{kT}\right) \tag{3.105}$$

Inserting ΔE_g from Eqs. (2.25) or (2.26) it is seen, that the h-value is strongly enhanced by the bandgap reduction. With increasing N_D the decrease of the first term holding for $\Delta E_g = 0$ will be overcompensated at higher N_D by the increase of the ΔE_g-term (see the later Fig. 3.21).

As mentioned, the highly doped region was assumed till now thick enough that recombination at the contact could be neglected. Often however the contact recombination is considerable or made large even intentionally. Fortunately, it can be taken into account simply by replacing the diffusion length L_p in (3.105) by a length $L_{p,eff}$ defined suitably for this purpose. Assuming a contact with infinite recombination velocity the minority carrier concentration at the surface is pressed down to the equilibrium value p_{n0} so that $p \approx p - p_{n0} = 0$. With this boundary condition at $x = 0$ the solution of the differential Eq. (3.42) is $p_n(x) = p_n^* \cdot \sinh(x/L_p)/\sinh(w_n/L_p)$ (w_n width of the emitter). The gradient of this carrier distribution at $x = w_n$ is

$$dp/dx(x_n) = p_n^*/(L_p \cdot \tanh(w_n/L_p)$$

which for $w_n \rightarrow \infty$ tends to the previous form p_n^*/L_p. Hence for small w_n the length to be used instead of L_p is

$$L_{p,eff} = L_p \cdot \tanh(w_n/L_p) \tag{3.106}$$

Thus the emitter parameter is finally given in this model by:

$$h_n = \frac{D_p}{N_D \cdot L_p \cdot \tanh(w_n/L_p)} \cdot e^{\Delta E_g/kT} \tag{3.107}$$

where the dependency on the doping and carrier concentration $N_D = n^+$ is contained apart from N_D itself in $L_p = \sqrt{(D_p \cdot \tau_p)}$, ΔE_g, and D_p. For very high N_D, where τ_p is determined by Auger recombination and $L_p = \sqrt{(D_p/c_{A,n})}/N_D$ is smaller than about $w_n/3$, Eq. (3.107) yields instead of (3.101)

$$h_n = \sqrt{D_p \cdot c_{A,n}} \cdot e^{\Delta E_g/kT} \tag{3.108}$$

Formulae for a p-emitter are obtained by exchanging n and p and N_D and N_A. Equation (3.107) for a p-emitter reads

$$h_p = \frac{D_n}{N_A \cdot L_n \cdot \tanh(w_p/L_n)} \cdot e^{\Delta E_g/kT} \tag{3.109}$$

For later use, we note also the formula for a small width of the p^+-region. If $w_p < L_n/3$, Eq. (3.109) turns into

$$h_p = \frac{D_n}{N_A \cdot w_p} \cdot e^{\Delta E_g/kT} \tag{3.110}$$

This equation holds also for position dependent emitter doping if $N_A \cdot w_p$ is replaced by the integral $\int N_A(x) \cdot dx$. For ΔE_g and D_n one has to use then suitable mean values. In Fig. 3.21 the h-parameter of a p^+-region as calculated from (3.109) is plotted versus the doping density N_A for a very small and a larger width w_p. The minority carrier lifetime τ_n was calculated from the Shockley-Read-Hall lifetime τ_{SRH} assumed to be 2 µs and Auger recombination according to $1/\tau_n = 1/\tau_{SRH} + c_{A,p} \cdot N_A^2$, where the Auger constant $c_{A,p}$ of (2.57) was inserted. As band gap narrowing $\Delta E_g(N_A)$ Eq. (2.25) of [Slo76] as well as Eq. (2.26) of Lanyon and Tuft [Lan79] were used. The minority diffusion constant $D_n(N_A)$ was calculated from the mobility μ_n using for simplicity the same dependence on doping concentration as in an n^+-region: $D_n(N_A) = kT/q \cdot \mu_n(N_D = N_A)$ (see Sect. 2.6 and Appendix A). As expected from Fig. 2.11, the ΔE_g of [Lan79] results in considerably smaller h_p below 10^{19} cm^{-3} than that of Slotboom and De Graaf. Experimental h-values obtained with power diodes are mostly in rough agreement with the $\Delta E_g(N)$-dependency of [Slo76]. At small N_A corresponding curves tend to the same value, since ΔE_g then approaches

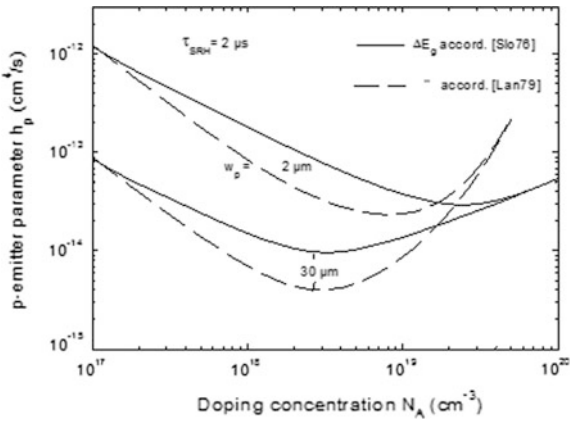


Fig. 3.21 h-parameter of a p^+-region as a function of doping concentration according to Eq. (3.109)

zero in both cases. As is shown already by the formulae, h can be strongly enhanced by choosing a small thickness of the emitter. The curves for $w_p = 2$ µm are described up to about 1×10^{19} cm^{-3} by (3.110) showing that recombination at the contact predominates. For $w_n = 30$ µm on the other hand, recombination in the emitter volume is significant in the whole range. Due to the decrease of the lifetime because of Auger recombination adding to the increase of ΔE_g with N_A, h_p has a minimum at 3×10^{18} cm^{-3}.

Concerning n-emitter regions, it can be seen, that due to smaller minority diffusion constant D_p and the higher Auger recombination constant $c_{A,n}$ the numerical h_n-values will be somewhat smaller than h_p under the same conditions. For a small emitter thickness the formula corresponding to (3.110) yields a h_n which is by the factor D_p/D_n smaller than h_p provided the integrated doping density is equal in both cases. The band gap narrowing is generally assumed to be the same in n^+ and p^+ regions. At very high doping densities the parameters h_n and h_p are very similar according to (3.108) since the products $D_n \cdot c_{A,p}$ and $D_p \cdot c_{A,n}$ differ not much. The bandgap narrowing is generally assumed to be the same in n^+ and p^+-regions

Although not all these consequences of the theoretical model have been verified precisely and the band gap narrowing itself is not yet completely understood quantitatively, the model has proved to be quite useful for describing the main influence of emitter layout on device behavior.

For the experimental determination of emitter properties, diodes with a p^+nn^+- or p^+pn^+-structure have been used in a forward current range where the injection levels in the base region is high. Hence (3.97) was used in the form

$$j_p(x_n) = q \cdot h_n \cdot n_B^{*2} \tag{3.111}$$

Typical highly doped emitter regions have h-values in the range 1×10^{-14} – 3×10^{-14} cm^4/s, and the lowest values observed are about 7×10^{-15} cm^4/s [Sco69, Bur75, Sco79, Coo83]. As will be treated in detail in Chap. 5, the h-values of the two emitter regions of pin-diodes and thyristors influence largely their forward voltage drop from moderate current densities upwards. For low-frequency power diodes which are designed to show a forward voltage drop as low as possible, small h-values are required. The injection efficiency of the junctions then is high. If h = 1×10^{-14} cm^4/s and $n_B^* = p_B^* = 1 \times 10^{17}/\text{cm}^3$ at a current density of 200 A/cm^2, one obtains from (3.100) $\gamma = 0.92$. On the other hand, the h-value of the p-emitter of fast power diodes and IGBTs is made often very high by choosing a small thickness and a relatively low doping density. In a typical example one may have $h_p = 1 \times 10^{-12}$ cm^4/s, and the carrier concentration in the base region at the p-emitter side may be 1.8×10^{16} cm^{-3} at a forward current density of 150 A/cm^2. In this case (3.100) yields $\gamma = 0.65$. The electron particle current which enters the p^+-region and ends at the contact by recombination, amounts then to more than a third of the total current.

An observation which is not expected from the above theory is that the minority carrier current flowing into the emitter regions contains besides the quadratic term

(3.111) also a significant part which increases proportional with the concentration n_B* [Sco79, Coo83]. Hence the minority carrier current density is actually given by

$$j_p(x_n) = q \cdot (h_n \cdot n_B^{*\,2} + s_n \cdot n_B^*) \tag{3.112}$$

This was found for all samples investigated regardless whether prepared by diffusion, epitaxy or alloying. The linear term in (3.112) is often higher than the recombination in the volume of the base. For gold doped pin-diodes with diffused junctions, Cooper [Coo83] found that the linear constant s_n in (3.112) is nearly proportional to the gold concentration. The linear recombination is caused possibly by an accumulation of recombination centers at the junctions. Its existence follows because the high-level lifetime in the base region determined from the carrier distribution is appreciably higher than the total linear recombination lifetime determined from stored charge measurements. See also Sect. 5.5.

3.5 Capacitance of pn-Junctions

In this section, we consider the capacitance of a pn-junction for small oscillations around a given stationary state at reverse bias or a small positive bias. The actual switching behavior of power devices consisting of large current and voltage changes between forward and reverse states will be discussed in later chapters. The small-signal capacitance in conjunction with an external stray inductance can be a source of disturbing oscillations in power electronic circuits (see Chap. 13). On the other hand, the capacitance is used as a tool for determining the doping concentration of the weakly doped base region of a junction. It is also used in different methods to investigate the properties of deep-level impurities which will be discussed in Sect. 4.9. In the present Section, however, the concentration of deep impurities is neglected against the density of the normal doping. The doping atoms are assumed to be completely ionized.

In a reverse biased pn-junction, the capacitance per unit area is defined as $c_j = dQ/dV_r$, where dQ is the variation of charge per unit area caused by an incremental change dV_r of the reverse voltage. Assuming a p^+n-junction the positive charge in depletion approximation , is given by $dQ = q \cdot N_D \cdot dw$, where N_D denotes the doping density of the base region and dw the change of the width w of the space charge layer caused by the voltage dV_r. The latter can be expressed according to Eq. (3.58) as $dV_r = q \cdot N_D \cdot w \cdot dw/\varepsilon$. This results in $c_j = dQ/dV_r = \varepsilon/w$, a well-known formula for a planar parallel plate capacitor. Substituting (3.58) for w, one obtains the capacitance per unit area as function of the reverse voltage:

$$c_j = \sqrt{\frac{q \cdot \varepsilon \cdot N_D}{2 \cdot (V_{bi} + V_r)}} \tag{3.113}$$

or

$$\frac{1}{c_j^2} = \frac{2 \cdot (V_{bi} + V_r)}{q \cdot \varepsilon \cdot N_D} \tag{3.114}$$

According to this equation, a plot of $1/c_j^2$ versus V_r yields a straight line, whose slope can be used to determine the doping concentration N_D. This measuring method has the advantage compared with other methods that it allows the direct determination of donor and acceptor doping concentrations down to 1×10^{13} cm^{-3}, without using any other quantities and material characteristics. Because the capacitance according to (3.113) refers to a depleted space charge region, it is called depletion capacitance.

If we assume for example $N_D = 1 \times 10^{14}$ cm^{-3}, $V_r = 600$ V and a junction area $A = 2$ cm^2, the capacitance $C_j = A \cdot c_j$ amounts to 235 pF. With a stray inductance $L_{par} = 50$ nH in series, this capacitance gives rise to oscillations with frequency $f = 1/(2\pi\sqrt{L_{par} \cdot C_j}) = 46.4$ MHz. Note that this far above the usual switching frequency in power electronic circuits.

As illustrated in Fig. 3.5, the space charge in abrupt asymmetrical junctions can be dominated by the charge of mobile carriers which is neglected in the depletion approximation for the capacitance (3.113). The exact expression for the maximum field strength in an abrupt junction is given by Eq. (3.24). Since from the Poisson equation the absolute charge of one sign in the space charge region is obtained as $Q = \int |\rho| dx = \varepsilon \mathsf{E}_m$, Q follows immediately from (3.24). Assuming again a p^+n junction with $N_A \gg N_D$, the exponential term in V_N can be neglected. Expressing furthermore the linear term in V_N according to (3.22) by V_P one obtains:

$$Q = \sqrt{2\varepsilon kTN_A \left[\exp\left(-\frac{qV_P}{kT}\right) + \frac{qV_P}{kT} - 1\right]} \tag{3.115}$$

The voltage V_P over the N_A region is given by Eq. (3.23), where the built-in voltage has to be replaced at reverse bias by $V_{bi} + V_r$. Hence, Q is given as an explicit function of V_r which by differentiation yields the capacitance. From (3.23) one obtains $dV_P/dV_r = N_D/N_A$ and $qV_P/kT = 1 + \chi$ where

$$\chi \equiv \frac{N_D}{N_A} \cdot \frac{q}{kT} \cdot (V_{bi} + V_r) \tag{3.116}$$

Since $q(V_{bi} + V_r)/kT \gg 1$, the variable χ can be comparable with 1, although $N_D/N_A \ll 1$. The differential capacitance per unit area is then obtained from (3.115) as

$$c_j = \frac{dQ}{dV_r} = \sqrt{\frac{q\varepsilon N_D}{2(V_{bi} + V_r)}} \cdot \frac{1 - e^{-1-\chi}}{\sqrt{1 + e^{-1-\chi}/\chi}} \tag{3.117}$$

c_j decreases with increasing N_A, i.e. decreasing χ. For $\chi > 2$, a not too small N_D/N_A and a sufficiently high reverse voltage V_r, only the depletion meaning expression of (3.113) remains. With decreasing χ the capacitance decreases significantly and for $\chi << 1$ it is given by

$$c_j = 1.042\sqrt{\chi} \cdot c_{j,dpl} \tag{3.118}$$

where $c_{j,dpl}$ is now the depletion capacitance per area as given by (3.113). At appropriate N_D/N_A *the capacitance of abrupt* p^+n and n^+p *junctions at zero bias is up to two orders of magnitude smaller than the depletion capacitance*. In Fig. 3.22 the inverse capacitance of p^+n junctions in Si is plotted versus the square root of $V_{bi} + V_r$ for $N_D = 1 \times 10^{14}$ cm^{-3} and some values of N_A. For $N_A/N_D \leq 10^3$ the dependency passes soon into the straight line depicted for the depletion capacitance. For $N_A = 1 \times 10^{18}$ cm^{-3} and the more so for 4×10^{18} cm^{-3} the voltage dependence is over a wide or the whole range significantly weaker than that of the depletion capacitance. The mentioned method of determining the doping concentration N_D from the voltage dependence of c_j is hence not applicable in these cases.

The condition of sharp abruptness assumed in the formulae is realized often by junctions in SiC and other wide-gap semiconductors, in silicon junctions made by low-temperature epitaxy are very abrupt. In most diffused junctions the doping transition is on the contrary slow enough that carriers from the highly doped zone do not form a considerable part of the space charge; hence the depletion approach

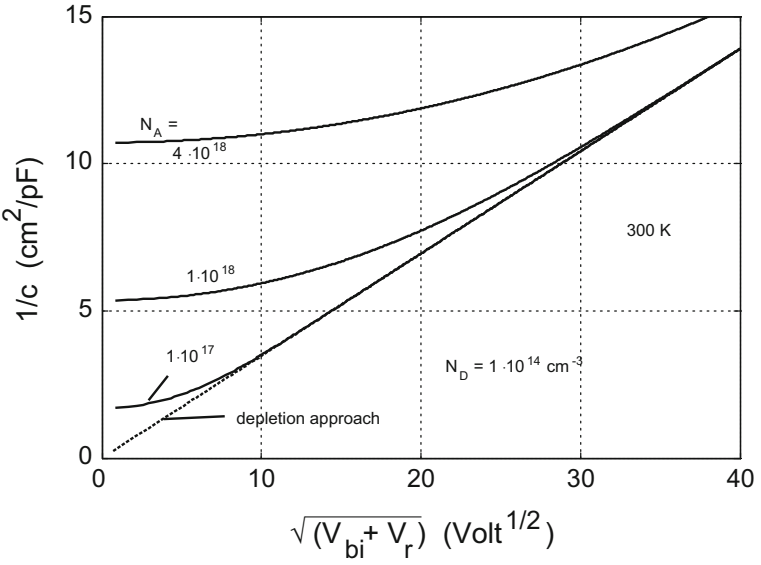


Fig. 3.22 Inverse capacitance of abrupt p^+n junctions in Si plotted versus square root of the voltage $V_{bi} + V_r$. The doping concentrations of the p^+ region is used as parameter

applies. Diffused junctions exhibit therefore a far higher capacitance than comparable abrupt high-low junctions, a noteworthy advantage of the latter.

Putting $V_r = -V_F$, the equations apply also to the capacitance at forward bias, if the voltage remains below several kT/q. At higher V_F the capacitance is dominated by a rearrangement of the minority carrier concentrations in the neutral regions. Concerning this capacitance contribution called 'diffusion capacitance' we refer to the literature [Sze02].

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